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TDK STUDY

From Senescence to Repopulation: Alternative Splicing Dynamics in Breast Cancer Cells

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2026.

Abstract

Alternative splicing (AS) is a major post-transcriptional regulatory mechanism that increases transcript diversity and influences a range of cellular processes, including proliferation, differentiation, and stress responses. Therapy-induced senescence (TIS) is a response to anticancer therapy characterized by prolonged growth arrest. Some senescent cancer cells can then escape senescence and re-acquire proliferative capacity, potentially leading to relapse. While some work has examined the transcriptional changes that occur during this process, the dynamics of alternative splicing during senescence and the subsequent repopulation phase remain poorly understood.

My aim was to characterize the alternative splicing changes that occur as breast cancer cells go from being proliferating to undergoing therapy-induced senescence and then to repopulation. I looked at publicly available bulk RNA-seq data from two breast cancer cell lines (MCF7 and T47D) across three conditions: untreated control (CTR), therapy-induced senescence (TIS), and repopulation (REPOP). Differential gene expression was assessed using RSEM and DESeq2, and alternative splicing was assessed using two different approaches (rMATS and LeafCutter), which were then combined to identify the robust splicing changes and to characterize the dynamics of those changes across the three cellular states.

Both cell lines showed extensive transcriptional changes during TIS, with strong suppression of cell cycle, DNA replication, and RNA processing pathways. Spliceosome-associated genes were suppressed, but the global distribution of splicing changes showed no systematic shift or increase in intron retention, indicating that the changes observed are not due to a generalized failure of the splicing machinery. Trajectory analysis revealed six distinct patterns of splicing dynamics, with transient and partially reversible changes being the most common. A significant proportion of changes remained after repopulation, and a further set of changes was observed specifically during the repopulation phase. The two methods showed 98% directional concordance on matched events, which confirms the validity of the changes identified. Importantly, only 7.1-14.7% of the differentially spliced genes were also differentially expressed, highlighting that alternative splicing reveals a range of molecular changes that standard gene expression analysis would miss. Changes in genes such as RPS24, ZNF207, and SRSF1 further illustrate this.

In summary, this work shows that alternative splicing undergoes widespread and heterogeneous reorganization during therapy-induced senescence and repopulation. The fact that some of the changes persist and new splicing patterns are acquired is strong evidence that repopulation is not simply a reversal of the senescent state. Alternative splicing is therefore a potentially important, but largely unexplored, layer of regulation in senescence escape and provides several candidate molecular changes for further study of the mechanisms underlying cancer cell recovery and relapse.

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1. Abbreviations

A3SS - Alternative 3' splice site

A5SS - Alternative 5' splice site

AS - Alternative splicing

BH – Benjamini-Hochberg

CTR – Control

DAS - Differential alternative-splicing

DGE - Differential gene-expression

ECM - Extracellular matrix

ER+ - Estrogen receptor-positive

FBS - Fetal bovine serum

FDR - False discovery rate

GO / GO_BP - Gene Ontology / Biological Process

GSEA - Gene-set enrichment analysis

KEGG - Kyoto Encyclopedia of Genes and Genomes

MSigDB - Molecular Signatures Database

MXE - Mutually exclusive exons

ORA - Over-representation analysis

PCA - Principal component analysis

PSI - Percent spliced in

QC - Quality control

REPOP – Repopulation

RI - Retained introns

ROS - Reactive oxygen species

RPM - Reads per million

RPMI - Roswell Park Memorial Institute

RT-PCR - Reverse transcription polymerase chain reaction

SA-β-gal - Senescence-associated beta-galactosidase

SAHF - Senescence-associated heterochromatin foci

SASP - Senescence-associated secretory phenotype

SE - Skipped exons

SRA - Sequence Read Archive

TIS - Therapy-induced senescence

2. Introduction

Cellular senescence is a state of stable cell cycle arrest in which the cell is metabolically active and will typically secrete a particular mixture of inflammatory and tissue remodelling factors. It is a potent barrier to tumor development but in the case of persistent senescent cells can also lead to inflammation and promote tumor progression so its effect on cancer is context-dependent. Senescence can be induced in cancer cells by both chemotherapy and radiotherapy. The senescence that is induced by these therapies (therapy-induced senescence or TIS) will limit the expansion of the cells that survive treatment, and for this reason it has been considered a desirable outcome of therapy.

The assumption that TIS is a permanent endpoint has recently been challenged. Cancer cells that have entered a senescence-like state can resume proliferation, a process termed senescence escape, which may contribute to tumor dormancy and relapse. Bajtai et al. showed that escape from TIS occurs in four breast cancer cell lines after high-dose chemotherapy, and that TIS provides a transient resistance to roughly half of the tested anticancer drugs. Bulk and single-cell RNA sequencing revealed that repopulating (REPOP) cells closely resemble the parental cells, whereas TIS cells differ markedly from both. TIS therefore appears to be a reversible cell state, but the molecular changes that accompany entry into this state and the return to proliferation are only partly known.

Differential gene-expression, which was the focus of that study, captures changes in the total amount of RNA from a gene. Alternative splicing adds a second regulatory layer, because the same gene can give rise to different transcript isoforms with different coding potential, stability or function. Isoform usage can therefore change when total gene expression does not, and analysing splicing can reveal regulation that gene-level analysis misses. Splicing is altered in cancer and in senescence. Many tumors show recurrent splicing changes, including widespread intron retention, and senescent cells show a general decrease in the expression of splicing factors and spliceosome components, accompanied by specific isoform changes.

However, most of this evidence comes from replicative or stress-induced senescence in non-transformed cells, mainly fibroblasts, and the data on TIS in cancer cells are limited to individual splicing factors. To my knowledge, the transcriptome-wide pattern of alternative splicing during the full course of TIS and repopulation has not been described.

In this work, I examined this question in the MCF7 and T47D breast cancer cell lines by analysing the published bulk RNA-seq data of control, TIS and repopulating cells. I identified differential alternative splicing events with two complementary methods, rMATS (event-level) and LeafCutter (intron-level), compared their results, and related the splicing changes to differential gene expression. I then followed individual events across the three cell states to determine which changes are transient and which persist after repopulation. The study is descriptive, and it generates hypotheses about splicing in senescence escape without testing the function of individual events.

3. Literature review

3.1 Cancer, aging, cellular senescence

3.1.1 The hallmarks of aging and their link to cancer

Aging is a complex biological process characterized by a step-by-step reduction in the body's physiological integrity and its capacity to maintain cellular and tissue balance. It results not from a single molecular defect but from the gradual accumulation and interaction of multiple molecular and cellular changes. To identify the main processes driving this decline, researchers introduced the concept of the hallmarks of aging. The first version of this framework identified nine hallmarks: genomic instability, telomere attrition, epigenetic alterations, loss of proteostasis, deregulated nutrient sensing, mitochondrial dysfunction, cellular senescence, stem cell exhaustion, and altered intercellular communication [1]. This group of processes provides a useful means of understanding aging since it covers the different types that occur, ranging from changes to the genome and modifications in cellular metabolism to changes in tissue maintenance and intercellular signalling.

The different aspects of aging should not be seen as distinct mechanisms. In order to expand the model, López-Otín and his team identified disabled macroautophagy, chronic inflammation and dysbiosis as additional processes [2]. The authors particularly emphasized that the hallmarks are interconnected: a change in one process can have an effect on many of the others, and it is together through their actions that the gradual loss of homeostasis associated with aging occurs. For example, genomic instability and telomere dysfunction can cause cellular stress responses, while mitochondrial dysfunction and altered intercellular communication can contribute to inflammatory signalling. Cellular senescence - a stress-induced state of stable cell-cycle arrest - is therefore not merely an isolated feature of aging but rather one part of a broader network of interconnected processes.

In this case the link between aging and cancer is especially important. As aging is the primary risk factor for a number of human diseases, including cancer, the rate of cancer generally increases with advancing age [1,3]. The biological changes that accumulate during the aging process can contribute to both the development and progression of cancer; genomic instability increases the probability of acquiring alterations that affect cellular proliferation and survival, and epigenetic changes can modify the regulation of the genes involved in these processes. Furthermore, the age-related changes in tissue homeostasis, stem cell function and intercellular communication can also influence the cellular environment in which malignant transformation occurs [1,2]. Therefore, the connection between aging and cancer cannot be merely attributed to the accumulation of mutations in individual cells, since cancer also develops in an aging tissue environment whose features change gradually over time.

At the same time a major biological paradox exists concerning aging and cancer. Even though various processes which protect tissues from becoming malignant may cause age-related dysfunction if they go on or become dysregulated, cellular senescence is a clear example of this apparent contradiction. It can prevent damaged or possibly malignant cells from continuing to divide and so serve as a barrier to tumor development; on the other hand, the accumulation of senescent cells and the effects they have on the surrounding tissue can also result in chronic inflammation, tissue dysfunction and, in certain cases, in tumor progression [3,4]. Cellular senescence therefore forms at the centre of the connection between aging and cancer.

3.1.2 Cellular senescence

The concept of cellular senescence arose from the observation that normal human cells have a limited capacity for division when cultivated in culture. Hayflick and his team demonstrated that human diploid cells do not divide without limit and instead eventually get to a stage at which further divisions cease. They called this the Hayflick limit, thus providing an initial experimental basis for the study of cellular ageing and replicative senescence [4]. Subsequent research has shown that a loss of proliferative ability is not restricted to cells that have reached a replicative limit; different kinds of cellular stress, such as DNA damage and oncogenic signalling, can also cause a senescence-associated state.

Historically, cellular senescence was defined as a strictly permanent withdrawal from the cell cycle. However, current evidence suggests this arrest is highly stable rather than absolutely irreversible, with cells remaining metabolically active. Instead of simply becoming inactive or dying, senescent cells undergo substantial changes in their molecular and cellular properties. The onset and maintenance of the growth arrest linked to senescence are based on major tumor-suppressive pathways, for example the p53–p21 and p16INK4a–Rb pathways. These pathways respond to cellular stress by stopping progression through the cell cycle and therefore prevent damaged cells from continuing to proliferate. The capacity of oncogenic signaling to induce senescence is particularly important when studying cancer: excessive activation of oncogenes can cause a cellular stress response which results in a stable growth arrest rather than uncontrolled proliferation [3].

The ability of senescence to suppress tumor formation helps to account for its evolutionary and physiological significance. If cells suffer damage at levels that could be harmful or if they receive oncogenic signals, they can be stopped from taking part in malignant transformation by entering a state of senescence. Senescence therefore functions as a cellular barrier against cancer. This is backed by the fact that senescent cells are found in pre-malignant tissues and that either their elimination or the failure to keep an eye on them has an effect on tumor development [5]. Senescence is thus one of the ways in which cells respond to potentially oncogenic stress before uncontrolled proliferation occurs.

The consequences of senescence are not confined to the senescent cell alone. Such cells can acquire a senescence-associated secretory phenotype (SASP), characterized by the production and secretion of a wide range of signalling molecules, including inflammatory cytokines, chemokines, growth factors and substances that alter the extracellular matrix [6]. Through these secreted factors, senescent cells are capable of influencing adjacent cells and the surrounding tissue microenvironment. Although the SASP may contribute to immune surveillance and tissue remodelling, persistent SASP activity can also cause chronic inflammation and, in some cases, create a microenvironment that is favourable for tumor progression [6,7].

The role of senescence in cancer is inherently paradoxical. Although senescence can inhibit the development of cancer by preventing the proliferation of damaged cells or cells in which oncogenes are activated, long-lasting senescent cells can have effects that go beyond their own cell-autonomous growth arrest by influencing neighbouring cells and the tissue environment via the factors that they secrete. Consequently, the biological result of senescence depends not just on the presence of growth arrest but also on the duration of the senescent state and on its functional effects [6,7].

The relationship between senescence and cancer may therefore be regarded as depending on the context rather than being simply tumor-suppressive or tumor-promoting: on the one hand, senescence can function as a natural obstacle to tumor development; on the other hand, persisting senescent cells may acquire or promote features that support tumor progression in particular biological contexts [7]. This apparent contradiction is particularly relevant in the case of cancer since cellular stress can be deliberately induced as part of treatment; a therapy that successfully causes tumor cells to enter senescence may at first halt their proliferation, but the long-term result will depend on the continued presence and subsequent behaviour of the senescent cells.

It is important to maintain this balance since cellular senescence has now become a potential target for therapy. Recent studies have looked into both the elimination of senescent cells and changes to their phenotypes, as senescence can be either beneficial or harmful depending on the biological context [7,8]. The fact that there is so much interest in therapeutic approaches indicates that senescence should not be viewed simply as an irreversible end point but should instead be regarded as a complex biological cell state whose results depend on the characteristics of the cells and their environment.

Aging, cancer, and cellular senescence are linked through a network of related molecular and cellular processes. Because of aging, changes progressively accumulate at the genomic, epigenetic, metabolic, and tissue levels, and the risk of cancer increases as these alterations build up. Cellular senescence is important in connecting these processes since it can help to stop malignant transformation by limiting the proliferation of damaged cells, but when senescent cells remain, they can alter the tissue environment and cause age-related dysfunction as well as promote tumor development. This dual role provides the biological

basis for considering cellular senescence not as a single endpoint but as a cellular state whose molecular features and effects can differ over time and in response to various types of stress.

3.2 Cellular senescence as a heterogeneous state

Cellular senescence is defined as a stable cell cycle arrest but is by no means a single, monolithic cellular programme. The senescent phenotype is highly heterogeneous, and the outcome depends on several factors, including the nature of the inducing stimulus, the cell type, the genetic background, and the microenvironment [9,10]. Different stressors (e.g. telomere shortening, persistent DNA damage, oncogene activation, oxidative stress) will activate different upstream signalling pathways even if they all ultimately result in cell cycle arrest [11]. These pathways may converge on common downstream regulators (in particular the p53-p21 and p16-RB axes), but each will contribute differently across contexts [9,11]. The same senescent phenotype can therefore be induced via different mechanisms. Cell identity will also play a major role in the response, as different cell types will have very different transcriptional programmes, metabolic states, chromatin states and signalling networks. The same stimulus may therefore lead to very different outcomes in different cell types, and different stimuli may produce overlapping phenotypes in cells with similar regulatory properties [9,10]. The functional state of tumor suppressor pathways and the microenvironment (neighbouring cells, the extracellular matrix, and soluble factors) also influence the establishment and maintenance of senescence [10–12].

There is clearly considerable heterogeneity at all levels. The cells are often enlarged and flattened, but this is very much cell type and condition-dependent and is not in itself a reliable marker of senescence [9,10]. The molecular phenotype is also highly model-dependent. The degree of cell cycle arrest, DNA damage signalling, the inflammatory response, metabolic changes, and stress response activation all vary depending on the stimulus used to induce senescence and the cellular background [11,13]. Chromatin organization is also very variable, with some cells forming the classic senescence-associated heterochromatin foci (SAHF) and others undergoing a different type of chromatin remodelling [11]. Mitochondrial function, energy metabolism, and ROS production also vary widely [11]. Crucially, these are interlinked rather than discrete features, as changes in chromatin structure affect gene expression, metabolic changes affect signalling pathways, and transcriptional changes in turn determine the profile of secreted factors [9,10].

The sheer diversity of the senescent phenotype makes it difficult to identify, with no single marker definitive of senescence. Common markers include SA-β-gal activity, p16 and p21 expression, persistent DNA damage, morphological changes, and the SASP. Each has drawbacks: p16 and p21 expression is highly context-dependent, SA-β-gal activity is not specific to senescent cells, and the mere presence of inflammatory factors does not equate to senescence [10]. Gorgoulis et al. therefore advocate using multiple independent markers to achieve a more robust assessment of the senescent state [10]. This heterogeneity has

significant implications for transcriptomic studies, as molecular features identified in one cell type/model may not apply to other cell types or experimental settings. Therefore, it is essential to consider both the commonalities and the context-specific features when examining the molecular regulation of senescence.

3.3 Therapy-induced senescence and senescence escape

3.3.1 Therapy-induced senescence in cancer

Therapy-induced senescence (TIS) is a cellular response to anti-cancer therapy in which cancer cells enter a state of prolonged growth arrest while remaining viable. Chemotherapy and radiotherapy are both potent inducers of TIS. Rather than dying immediately, some treated cells survive and acquire the hallmarks of senescence (morphological, signalling, and gene expression changes). TIS is therefore a distinct cellular response to therapeutic stress, separate from apoptosis and other forms of cell death [14].

Senescence has long been viewed as a potentially beneficial outcome of cancer therapy. By preventing cell division, it limits the expansion of cancer cells that survive therapy. This is particularly relevant when therapy fails to eradicate the whole tumor, as durable growth arrest in the remaining cancer cells could provide an additional tumor-suppressive mechanism beyond direct cell killing [15,16].

However, the long-term survival of senescent cancer cells complicates this picture. Growth arrest does not necessarily equate to an irreversible endpoint, and it is entirely possible that some cells will eventually regain their proliferative capacity. The therapeutic efficacy of TIS will therefore depend not just on the induction of growth arrest but on the stability of that arrest and the ultimate fate of the cells that survive it [14,16].

3.3.2 TIS as a transient drug-resistance state

This is central to Bajtai et al. (2025), who looked at therapy-induced senescence as a potential transient mechanism of drug resistance in breast cancer [15]. They looked at four breast cancer cell lines after high-dose chemo and followed their recovery. They found that TIS is not permanent, showing that senescent cancer cells can survive therapy and then re-enter the cell cycle [15].

The surviving cells also showed altered sensitivity to subsequent anti-cancer therapies. This differs from the classic scenario of drug resistance, in which cancer cells continue to proliferate despite treatment. In senescence, cells survive but do not proliferate and may later re-enter the cell cycle. Bajtai et al. treated senescent cells with a range of approved anti-cancer drugs of different modes of action and found resistance to roughly half of them. This indicates that the altered response to therapy is not simply a consequence of resistance to a single drug target but is likely to be a feature of the senescent state per se [14,15].

The authors then compared parental, TIS, and repopulating cells using molecular profiling. TIS cells differed markedly from both parental and repopulating cells, whereas parental and repopulating cells were quite similar in their gene expression profiles [15]. This indicates that many of the transcriptional changes associated with senescence are not necessarily 'locked in' following proliferative recovery. That said, similarity in gene expression does not equate to molecular identity, and it is not clear how much of the individual changes are retained following repopulation. The study also underlines the potential biological relevance of the senescent cells that survive treatment. They may be a residual population capable of contributing to subsequent tumor growth despite being growth-arrested initially.

3.3.3 Senescence escape and repopulation

Senescence escape is the phenomenon of cells regaining proliferative capacity after entering a senescent state. It directly contradicts the long-held view of senescence as an irreversible cell-cycle arrest. Senescence escape has been observed in a number of cancer models following treatment with doxorubicin, etoposide and camptothecin (among others) [14,17,18]. The extent and nature of escape vary dramatically from one system to another.

Early evidence came from Roberson et al. (2005), who examined escape from therapy-induced accelerated senescence in p53-deficient lung cancer cells. They observed the re-establishment of proliferation in their models and linked alterations in cell cycle regulation (particularly cyclin-dependent kinase activity) to this [17]. Later work has shown that tumor cell populations enriched for senescent characteristics can regain proliferative capacity and, under the right experimental conditions, form tumors after transplantation [18]. Saleh et al. (2019) discussed a potential link between senescence escape, tumor dormancy, and recurrence, suggesting that a residual population of tumor cells could survive therapy in a state of prolonged growth arrest before a subpopulation regains proliferative capacity [16]. That said, senescence is only one of the possible forms of tumor dormancy and escape from senescence cannot be the explanation for all cases of cancer recurrence.

An important experimental issue is distinguishing true escape from senescence from the mere proliferation of cells that never entered senescence. After chemo, you can end up with a population of cells that have undergone apoptosis, some that remain proliferative, some that have entered a long period of growth arrest, and some that have then re-entered the cell cycle. Population regrowth alone is therefore not sufficient to demonstrate escape from senescence; you need evidence of the prior senescent state and subsequent recovery [14,18]. In this respect, the longitudinal analysis of Bajtai et al. provides evidence for delayed repopulation following chemotherapy-induced senescence in breast cancer models and thus supports the idea of TIS as a potentially transient cellular state [15].

3.3.4 Molecular changes associated with senescence escape

The molecular mechanisms that underpin escape from senescence are still not fully understood. Because senescence results from the interplay of cell cycle, stress response, and survival pathways, it is highly unlikely that a single universal mechanism governs the ability to re-enter the cell cycle. The mechanism(s) involved are likely to be different depending on the genetic background of the cells and the nature of the stimulus that led to the induction of senescence [14,17]. Cell cycle regulation is one potential control point. The p53-p21 and p16-RB pathways are the principal mediators of growth arrest associated with senescence, and their status strongly affects the cell's response to stress. For instance, changes in cyclin-dependent kinase activity have been linked to escape in p53-deficient lung cancer cells [17]. That said, these findings should not be extrapolated to all cancer models.

Bajtai et al. examined several well-established mechanisms of drug resistance and found that none fully accounted for the particular resistance profile of TIS cells. Their molecular analyses did, however, identify changes in gene expression consistent with immune evasion and suggested a possible role for KRAS-associated signalling in escape from senescence [15]. This points to other regulatory mechanisms that may be involved in the survival and recovery of senescent cancer cells, although their precise functional roles remain to be determined.

Comparing parental, TIS, and repopulating cells is useful for distinguishing changes associated with the senescent state from those present after proliferative recovery. The large differences between TIS and the other two populations clearly indicate that senescence is associated with massive molecular reprogramming, much of which is reversed during the repopulation phase [15]. The extent to which the various regulatory layers recover in concert remains unknown. Changes in transcript abundance provide only a partial picture of the cell's molecular state.

Alternative splicing is another level of RNA regulation whereby a single gene can give rise to multiple transcript isoforms that may, in turn, affect RNA stability, translation, and protein function. Splicing changes could therefore reveal molecular differences between parental, senescent, and repopulating cells that would not be apparent from a standard gene-level expression analysis. However, splicing dynamics during therapy-induced senescence and the subsequent repopulation phase are extremely poorly characterized. Looking at splicing changes alongside differential gene expression may therefore provide a more complete picture of the transcriptomic reprogramming associated with senescence escape.

3.4 Alternative splicing as a layer of regulation

3.4.1 Molecular mechanisms and types of alternative splicing

The amount of RNA transcribed from a gene does not always decide its final functional outcome. Before it can be translated, pre-mRNA undergoes through a number of processing steps, splicing being one of the most important. This process involves the removal

of the introns and the joining together of the remaining exonic sections. In constitutive splicing a single mature transcript is produced, while alternative splicing allows for the inclusion or exclusion of certain exons or parts of the transcript. As a result, one gene is able to produce a number of different transcript isoforms which may vary in sequence and, in some instances, in function. Therefore, splicing is a major regulatory stage between transcription and the gene's eventual activity [19–22].

Alternative splicing affects the vast majority of human multi-exon genes [23, 24], and its outcomes are usually classified into a small number of basic event types, depending on which part of the pre-mRNA is retained in or removed from the mature transcript. The commonly distinguished classes are cassette (skipped) exons, alternative 5′ and 3′ splice sites, mutually exclusive exons and intron retention, with alternative promoter usage and alternative polyadenylation sometimes counted as further classes [25]. The first five can be quantified directly from exon-coverage and splice-junction data and are the event types analyzed in this study (Figure 1).

In exon skipping (SE), a cassette exon is either included between its two flanking exons or skipped, in which case the upstream exon is joined directly to the downstream one. In alternative 5′ splice site (A5SS) and alternative 3′ splice site (A3SS) events, competing donor or acceptor sites within the same exon lead to a longer or shorter exon version. In mutually exclusive exons (MXE), exactly one of two alternative exons is included, never both. In intron retention (RI), an intron remains in the mature transcript instead of being excised.

Intron retention deserves particular attention, since it is the least intuitive class and is central to the senescence and cancer literature discussed above. It is often overlooked because it is interpreted as a splicing mistake, yet there is evidence that intron retention is regulated [26]. Retained introns frequently introduce premature stop codons or keep the transcript in the nucleus, so that intron retention can lower the amount of productive mRNA and thereby tune gene expression [27]. The functional consequence of the other event types depends on the exon in question: an in-frame exon can add or remove a protein domain, whereas a frame-shifting exon can target the transcript to RNA surveillance pathways [22]. An event type alone therefore says little about its biological effect, and this must be assessed for each event separately

Differences between the various transcript isoforms can lead to important functional variations. Since alternative transcripts can give rise to proteins that have different domains, cellular localizations, stabilities, or interaction partners, the effects of splicing go beyond the protein-coding region, as splicing can also affect the destiny of the RNA molecule. Some of the isoforms result in stable, protein-coding transcripts, whereas others are kept in the nucleus or are directed towards degradation by RNA surveillance mechanisms. Thus, splicing not only determines which transcripts are produced but also influences their availability for translation and the quantity of functional protein that is generated. Moreover, the timing of splicing is significant. Intron removal does not take place at a fixed stage following transcription, and the

timing of processing can have an effect on RNA maturation, nuclear export, and protein synthesis [19, 21, 22].

Splicing is closely related to transcription and it is often possible to decide which splice sites to use while the pre-mRNA is still being produced. The speed at which RNA polymerase II travels along the gene, the recruitment of factors involved in RNA processing, and the chromatin environment surrounding the gene can all have an effect on the choice of splice sites. As a result, the final makeup of the different transcript isoforms depends not just on the sequence of the pre-mRNA but also on the conditions existing during transcription. Since transcription and splicing are therefore linked processes, a change in one of them can have an effect on the other and in the end lead to a different fate for the resulting RNA [21, 22].

Alternative splicing is mainly controlled by RNA-binding proteins and other splicing factors. These proteins bind to regulatory sequences that are found in both exons and introns, and the impact they have does so depending on both their activity and their position in relation to the splice sites. The functions of these factors may change as a result of different conditions in the cell. If the amount, location, or activity of particular splicing factors changes, this can at the same time affect a number of transcripts. As a result, changes in alternative splicing generally indicate wider alterations in cellular regulation rather than being merely isolated occurrences. This is especially important at times of major cellular transitions, such as during development, differentiation, or in disease [22,29,30].

Analysis of gene expression and splicing can provide different insights into the same biological system. Two different cellular states might have the same total level of RNA for a particular gene but could still give rise to quite different transcript isoforms. On the other hand, the amount of various isoforms can change even if the overall gene expression stays constant. While differential gene expression analysis mainly picks up on changes in the total amount of RNA, splicing analysis looks at the processing of that RNA. It therefore makes sense to combine both methods in order to obtain complementary views of the way in which gene regulation takes place during transitions between cellular states [19, 20, 22]

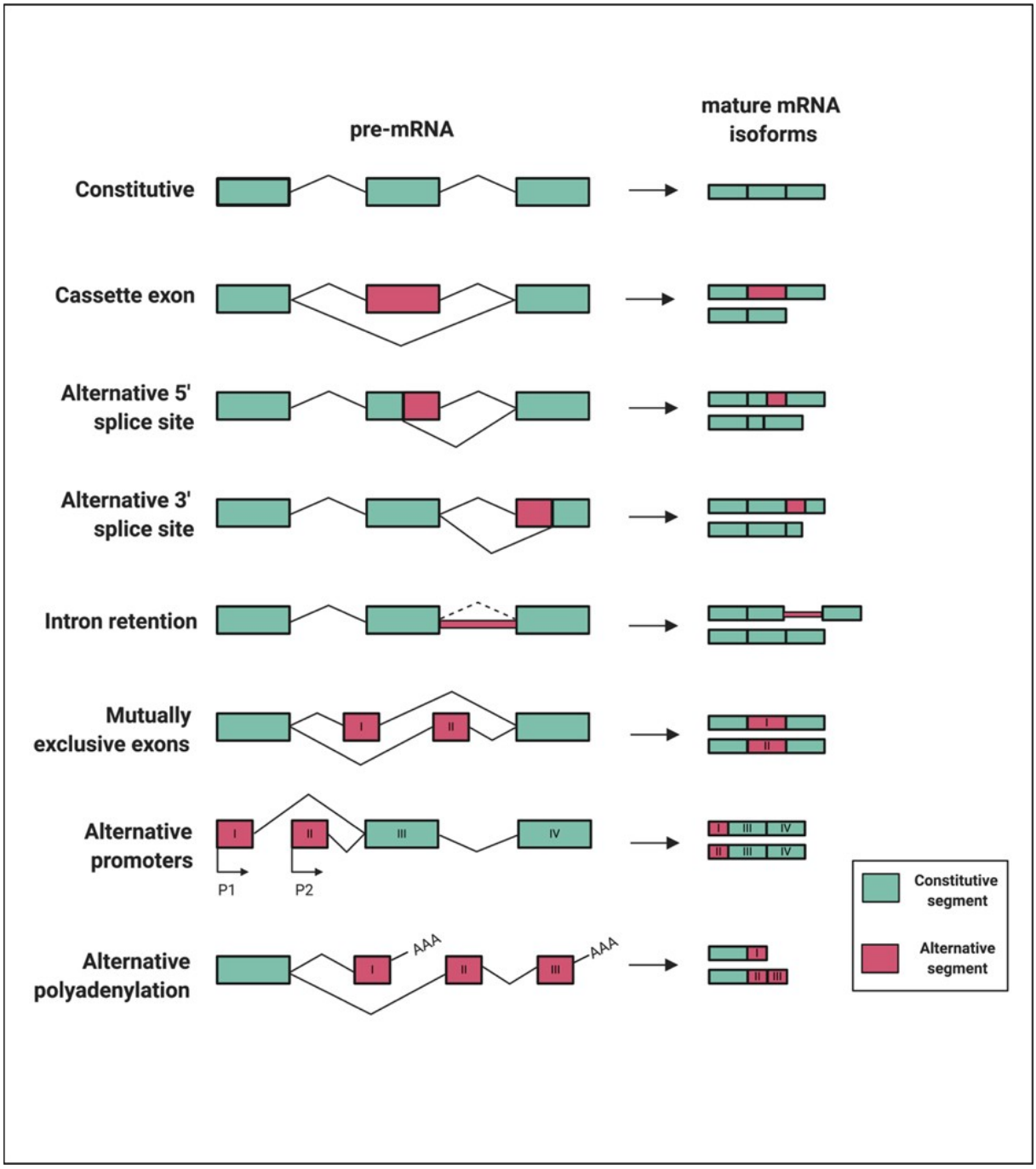


Figure 1. Mechanisms of alternative splicing producing mature mRNA isoforms [28].

It is particularly important to keep this distinction in mind when dealing with cellular transitions that are marked by widespread molecular reprogramming. Gene expression changes do not always lead to transcripts that are functionally equivalent to those which were present prior to the transition. Moreover, transcript processing can alter considerably even when there are only minor changes in the total level of gene expression. Therefore, by examining alternative splicing together with gene expression it is possible to detect regulatory changes that might be missed if one relies solely on gene-level measurements. In this way,

splicing offers an extra dimension to our understanding of gene regulation by linking changes at the transcriptional level to modifications in RNA composition, processing, and fate [19–22,29].

3.4.2 Quantifying splicing events

The usage of the alternative configurations of an event is expressed as percent spliced in (PSI, ψ), the proportion of transcripts that contain the included form among all transcripts covering the event. For a skipped exon, inclusion reads are those from the upstream junction, the alternative exon itself and the downstream junction, and skipping reads are those from the junction that directly connects the upstream and downstream exons [31]. Since the included form is longer and therefore attracts more reads for the same molecular abundance, the counts must be corrected for the length of the two forms. rMATS reports the lengths of the inclusion and skipping forms for this normalisation and calculates the inclusion levels from the normalized counts [31]. PSI therefore ranges from 0 (event fully skipped) to 1 (fully included), and the difference between two conditions (ΔPSI or dPSI) is expressed in absolute units, i.e. percentage points, and not as a relative change.

The meaning of “inclusion” is a convention fixed for each event type and has to be considered when interpreting the sign of ΔPSI. In rMATS, the inclusion form of an RI event is the one that retains the intron, that of an A5SS or A3SS event is the one that uses the long exon, and for MXE events it depends on the strand of the gene [31]. A positive ΔPSI thus means more exon inclusion for SE events but more intron retention for RI events.

3.5 Alternative splicing in cancer and cellular senescence

3.5.1 Splicing deregulation in cancer

Deregulated splicing is a well-established feature of cancer and can occur via a number of different mechanisms. It can result from recurrent somatic mutations in spliceosomal proteins, which alter splice site and exon recognition, and lead to cancer-specific mis-splicing. The most prominent examples are mutations in SF3B1, SRSF2, and U2AF1. Splicing defects in tumors can also result from mutations in the regulatory elements of individual cancer genes or from altered levels of splicing factors, with the resulting isoforms affecting cancer hallmark pathways. Splicing alterations are not limited to those tumors which have mutations in the spliceosome itself. A pan-cancer study found widespread intron retention in most cancer types without mutations in the spliceosome machinery (except breast cancer, where intron retention was a feature of adjacent normal tissue rather than the tumor). The direction and type of splicing change are therefore highly tissue- and context-dependent, and no single event class can be considered a universal cancer signature [26].

3.5.2 Splicing factors in senescence

Evidence linking the splicing machinery to senescence comes mostly from replicative or stress-induced senescence in non-transformed cells, and data on therapy-induced senescence in cancer cells are limited to individual factors. For example, SF3B4 expression decreases in replicative senescent fibroblasts, and its knockdown induces senescence through p21; in A549 lung cancer cells, SF3B4 overexpression reduces doxorubicin-induced senescence [25]. Spliceosomal components can therefore influence therapy-induced senescence, although this concerns a single factor rather than transcriptome-wide changes in isoform usage.

3.6 Bulk RNA-seq enables splicing analysis

RNA sequencing (RNA-seq) allows for the study of alternative splicing together with standard gene expression analysis. The sequencing fragments can be aligned to both exons and to the exon-exon splice junctions, thus providing quantitative data regarding the relative use of particular exons and transcript isoforms. By looking at how the RNA-seq fragments are distributed among alternative and constitutive regions it is possible to estimate the relative amount of alternatively spliced exons and isoforms, including the fraction of transcripts that contain a given alternative exon [32].

RNA-seq data can also be used to identify splicing events by means of reads or fragments that span splice junctions or include alternatively used exons. For example, reads that span the junctions between non-consecutive exons can serve as evidence for exon skipping, and reads that support particular exon–exon junctions or retained intronic regions can be used to quantify various kinds of alternative splicing events. Paired-end sequencing gives extra information by sequencing both ends of the same RNA fragment. The length of the fragment and the genomic positions of the paired reads can then be included in the estimates of isoform usage, thus providing further information for distinguishing between alternative transcript structures [32].

RNA-seq analyses are available at various levels of resolution. By looking at exon coverage and splice-junction data, it is possible to identify and measure individual splicing events such as exon skipping, intron retention, the use of alternative splice sites, and mutually exclusive exons. Instead, isoform-based methods aim to estimate the relative amount of each complete transcript isoform. Nevertheless, short-read RNA-seq usually gives only partial information regarding each transcript, so the full structure of an isoform cannot always be deduced directly from the individual reads. As a result, computational methods are needed in order to work out transcript structures or to quantify individual splicing events from the sequencing data that is available. A variety of computational methods have therefore been developed to detect and quantify differential alternative splicing from RNA-seq data, using measures based on exon usage, splice-junction usage, or specific splicing events [33].

In general, RNA-seq offers a method that allows for the study of changes in gene abundance as well as alterations in transcript processing over the whole transcriptome. It makes it possible to detect changes in exon or intron usage between different biological conditions and aids in the characterization of specific alternative splicing events linked to particular cellular states. Thus, splicing analysis supplements conventional differential gene expression analysis by supplying information on changes in transcript structure and isoform usage [32,33].

3.7 Objectives

Alternative splicing provides an additional layer of regulation which is not picked up at the gene level and has not been examined across the transition from proliferating cancer cells through to therapy induced senescence (TIS) and then repopulation. In the breast cancer model of Bajtai et al. [15] there are very significant differences in expression between the parental (CTR), the TIS and the repopulating (REPOP) cells and the senescent state is not maintained upon re-entry into proliferation. It is not known whether alternative splicing behaves in the same way.

The aim of the study was to characterize the changes in alternative splicing during TIS and repopulation in the MCF7 and T47D cell lines and to determine whether the splicing changes associated with TIS are reversed, maintained or newly acquired following proliferative recovery.

The specific objectives were:

1. To identify differential alternative splicing events (exon skipping, retained intron, alternative 5' and 3' splice sites, mutually exclusive exons) in the CTR-TIS, TIS-REPOP and CTR-REPOP comparisons using bulk RNA-seq and two complementary approaches: event-based (rMATS) and intron-based (LeafCutter) analysis.
2. To assess agreement between the two methods in the overlap of significant findings and the direction of splicing changes, and to compare the splicing landscape of the two cell lines.
3. To examine the relationship between splicing and gene expression, i.e., whether splicing changes occur alongside differential expression of the same genes or are independent of it.
4. To classify splicing changes by their trajectory through CTR, TIS, and REPOP (transient, reversed, partial, persistent, progressive, REPOP-specific) and test whether those trajectories are consistent across cell lines.
5. To identify well-supported events at individual loci (including genes involved in RNA processing) as candidates for functional follow-up.

The study is purely descriptive and does not set out to establish any causal role for individual splicing events in either the induction of senescence or in escape from senescence; any such conclusions would require functional validation.

4. Methods

The overall computational workflow, integrating gene expression and alternative splicing analyses from raw bulk RNA-seq data, is summarized in Figure 2.

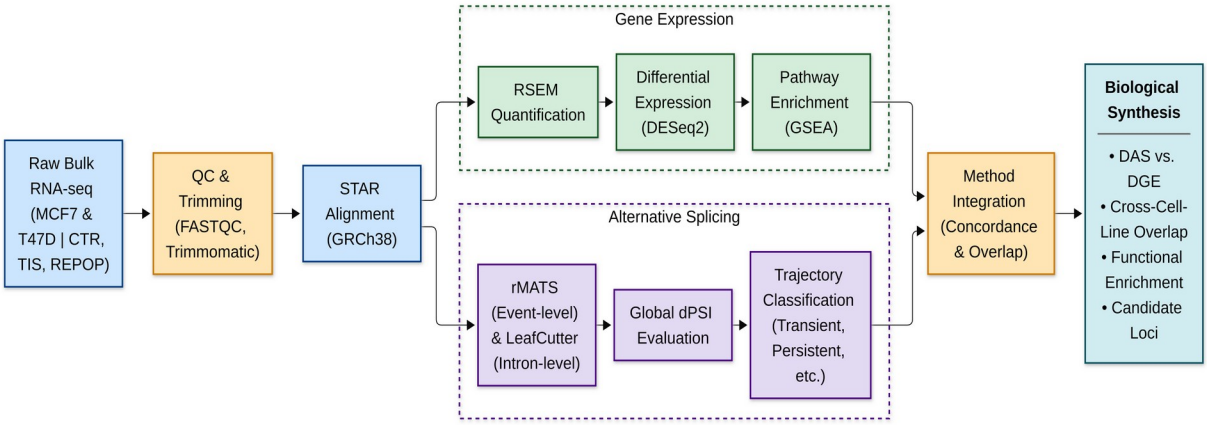


Figure 2. Schematic overview of the bioinformatic analysis pipeline, detailing the workflow from raw RNA-seq data preprocessing through parallel gene expression and alternative splicing streams to downstream integration and biological synthesis.

4.1 Experimental design and datasets

Bulk RNA-seq data from the ER+ (luminal) breast cancer cell lines MCF7 and T47D from Bajtai et al. [15] (GEO accession GSE287953) were used. Cells were grown in RPMI with 10% FBS at 37 °C and 5% CO2. Therapy-induced senescence (TIS) was induced with a high dose of doxorubicin (120 nM for MCF7, 70 nM for T47D), which killed over 90% of the cells. The cells were washed, fresh medium was added, and the surviving cells (approximately 7% of MCF7 and 8% of T47D) were maintained for 7 more days; TIS cells were harvested on day 12. A small proportion of TIS cells escaped the arrest and re-entered the cell cycle, giving rise to repopulating (REPOP) cells, harvested between days 24 and 36 at approximately 80% confluency. For each cell line, three conditions were analyzed: untreated parental cells (CTR), TIS and REPOP, with three biological replicates per condition. Total RNA was extracted with TRIzol and the Direct-zol kit (with on-column DNase I). Strand-specific libraries were prepared after poly(A) selection (NEXTFLEX Rapid Directional RNA-seq Kit 2.0) and sequenced on an Illumina NovaSeq 6000 (approx. 50 M 150 bp paired-end reads per library)

[15]. Raw reads were downloaded from the NCBI SRA and converted to FASTQ with the SRA Toolkit (3.4.1).

4.2 Preprocessing, alignment and quantification

Read quality was assessed with FastQC (v0.12.1) before and after trimming (one T47D control sample failed quality control standards and was excluded from subsequent analyses). Adapters and low-quality bases were removed with Trimmomatic (v0.41, ILLUMINACLIP:2:30:10 LEADING:3 TRAILING:3 SLIDINGWINDOW:4:15 MINLEN:36). Reads were aligned to the GRCh38 genome (GENCODE v45 annotation) with STAR (v2.7.11b) using the GTF annotation, producing sorted BAM files for the splicing analysis. Gene-level expression was quantified from the same reads with RSEM (v1.3.3). This pipeline is independent of the one used in the original study (STAR, featureCounts, edgeR and limma) [15].

4.3 Quality control and differential gene-expression

Sample-level similarity was assessed by principal component analysis (PCA) of variance-stabilised (vst) counts (using the top 500 most variable genes) and by Pearson correlation matrices between samples, separately for each cell line. RSEM expected counts were rounded to integers and analyzed per cell line with DESeq2 (v1.38.3) in R (v4.2.3). Samples that failed QC were removed, and genes with a total count below 5 x the number of samples were filtered out. The model was ~condition (CTR, TIS, REPOP). To obtain all three pairwise contrasts (CTR vs TIS, TIS vs REPOP, CTR vs REPOP), the model was fitted twice, with CTR and with REPOP as the reference level. Significance was assessed with the Wald test (alpha = 0.05, which also sets the independent-filtering threshold). log2 fold changes were shrunk with apeglm [34] and oriented as second minus first condition (TIS-CTR, REPOP-TIS, REPOP-CTR), in line with the dPSI convention below. A gene was called differentially expressed (DGE) if its Benjamini-Hochberg (BH) adjusted p-value was < 0.05 and its shrunken |log2 fold change| was > 1. Genes with a missing adjusted p-value (count outliers or independent filtering) were considered not tested. Counts shown in figures were normalized with DESeq2 size factors.

4.4 Gene-set enrichment analysis

GSEA was performed with clusterProfiler (v4.6.2) on genes ranked by shrunken log2 fold change for the CTR vs TIS and REPOP vs TIS comparisons, using KEGG pathways. Gene sets with BH-adjusted $p < 0.05$ were considered significant.

4.5 Differential splicing analysis

Event level differential splicing was assessed for each cell line using rMATS (v4.1.2) [31]. rMATS identifies skipped exons (SE), retained introns (RI), alternative 5' and 3' splice sites (A5SS, A3SS) and mutually exclusive exons (MXE), and calculates the inclusion level (PSI) of each event from the junction reads (JC output), normalized for the effective length of the inclusion and skipping forms. ΔPSI was defined as PSI(second condition) - PSI(first condition) (TIS-CTR, REPOP-TIS, REPOP-CTR) in accordance with the --b1/--b2 assignment. The analysis was performed in paired end mode (-t paired) with a read length of 150 bp (--readLength 150), taking into account variable read lengths (--variable-read-length), using the default statistical threshold (--cstat 0.0001) and restricted to annotated splice sites. Events were retained if the mean of the inclusion and skipping junction counts was at least 30 and were deemed significant at FDR < 0.05 and |ΔPSI| > 0.1. An event was defined as a unique combination of event type, chromosome, strand and genomic coordinates.

Intron-level differential usage was analyzed using LeafCutter [35]. Junction reads were extracted from the BAM files using bam2junc.sh (using regtools v1.0.0), and introns were clustered with leafcutter_cluster.py using a minimum of 50 reads per cluster (-m 50), the default minimum cluster ratio of 0.001 (-r 0.001), and a maximum intron length of 500,000 bp (-l 500000). Differential splicing across conditions was tested with leafcutter_ds.R using a minimum of 3 samples per group and 1 sample per intron. Introns were considered significant at FDR < 0.05 and |ΔPSI| > 0.1 (ΔPSI is the change in the fraction of cluster reads supporting the intron). Introns were assigned to genes by overlapping junction coordinates with GENCODE v45 annotations; for the cross-cell-line concordance analyses, T47D splicing events were assigned to genes by coordinate-based matching to the shared intron definitions identified in the MCF7 reference dataset and *vice versa*.

4.6 Global splicing landscape and cross-cell-line overlap

To test whether TIS causes a general shift in splicing, the distributions of ΔPSI (TIS−CTR) were plotted for all five event types in each cell line, with RI events inspected for a systematic positive shift. Overlap between MCF7 and T47D was quantified with the Jaccard index, $J = |A \cap B| / |A \cup B|$, where A and B are the sets of significant rMATS events (or, for LeafCutter, significant genes) in the two cell lines, each restricted to the common testable universe, i.e. events (genes) tested in both cell lines. Overlap is therefore not inflated by differences in detectability, and this value is referred to as the corrected Jaccard index. Results are reported in different units, which are never directly compared: rMATS results as events (an event significant in several comparisons is counted once when comparisons are pooled), and LeafCutter results as genes with at least one significant intron.

4.7 Classification of splicing trajectories across repopulation

To track the splicing dynamics from CTR through TIS to REPOP, two metrics were taken from the rMATS output for each event: ΔPSI_TIS (TIS vs CTR) and ΔPSI_REPOP (REPOP vs CTR). Events significant in CTR-TIS (FDR < 0.05, |ΔPSI| > 0.1) were then classified by the ratio r = ΔPSI_REPOP / ΔPSI_TIS:

- Reversed (r < -0.33): the change went in the opposite direction to the CTR baseline.
- Transient (−0.33 ≤ r < 0.33): the change was largely back to control levels.
- Partial (0.33 ≤ r < 0.67): the change partly reversed, in the original direction.
- Persistent (0.67 ≤ r ≤ 1.33): the change was largely maintained.
- Progressive (r > 1.33): the change continued in the same direction.

Events which were not significant in CTR-TIS but were significant in CTR-REPOP were classified as REPOP specific. Events which were significant in neither comparison were not classified. These classes are based on the ΔPSI point estimates; a non-significant difference in CTR-REPOP is indicative of the absence of a detectable difference (not proof of identity) given that there are three replicates per condition. Trajectory distributions were compared descriptively between MCF7 and T47D (Figure 3).

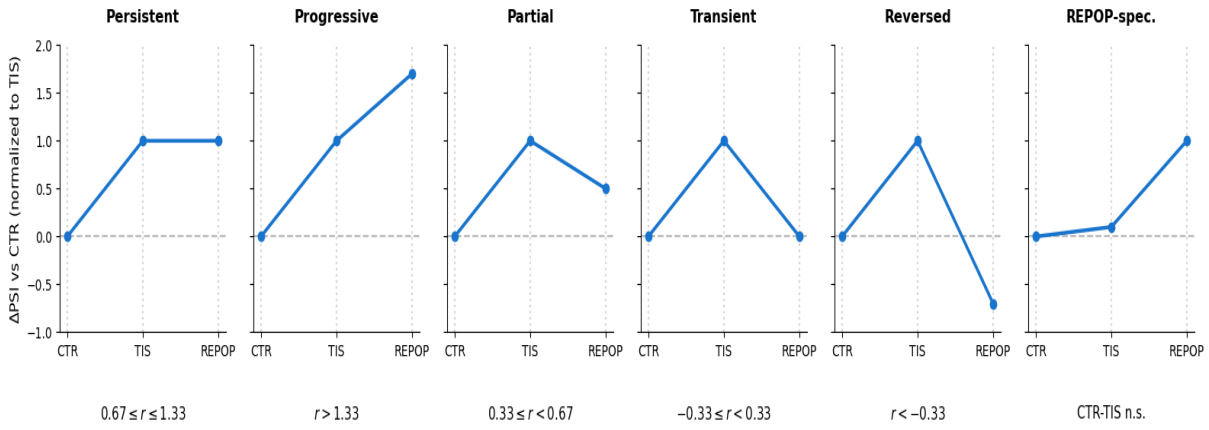


Figure 3. Splicing trajectory patterns across the control (CTR), therapy-induced senescence (TIS), and repopulation (REPOP) continuum. Schematic representation of alternative splicing dynamics throughout senescence induction and post-treatment recovery, normalized to TIS levels.

4.8 Integration of rMATS and LeafCutter results

As the two tools describe splicing differently, results were matched by genomic coordinates rather than by event ID. For SE events, the LeafCutter intron was matched to the skipping junction of the rMATS event (the intron had to begin immediately after the upstream exon end and end exactly at the downstream exon start, requiring an exact coordinate match

after chromosome and strand alignment). To make the effects comparable, the LeafCutter ΔPSI of the skipping intron was multiplied by −1 (intron excision and exon inclusion change in opposite directions). When several introns matched one event, their sign-harmonised ΔPSI values were averaged per cell line, comparison and event. Directional concordance was defined as the proportion of matched events whose two ΔPSI values had the same sign. To avoid circularity and selection bias, concordance was evaluated on significant rMATS events without applying any significance (FDR) or effect-size (|ΔPSI|) cutoffs to the LeafCutter output. At the gene level, a gene was supported by both tools (consensus) if it was significant in rMATS and in LeafCutter for the same comparison, not necessarily through the same event. Gene sets were compared across the six tool × comparison combinations in each cell line with Venn and UpSet plots.

4.9 Over-representation analysis

ORA of genes with shared splicing changes (i.e. the consensus DAS genes with significant alternative splicing in both MCF7 and T47D during TIS) was carried out using clusterProfiler (v4.6.2) against the MSigDB canonical pathway collection (CANONICAL_ALL includes REACTOME, WIKIPATHWAYS, KEGG_MEDICUS, BIOCARTA and PID), the Gene Ontology Biological Process (GO_BP) and the Hallmark gene sets via the msigdbr package. The background universe was all of the genes that were expressed in the splicing analysis and terms with BH-adjusted $p < 0.05$ were considered significant.

4.10 Relationship between splicing and gene expression

A gene was considered differentially alternatively spliced (DAS) if it was significant in both rMATS and LeafCutter for the same comparison (consensus definition, Section 4.8). The splicing results were combined with the DGE data by matching the Ensembl gene IDs (with the version suffixes removed). Gene symbols were then mapped to Ensembl IDs using the org.Hs.eg.db and EnsDb.Hsapiens.v86 annotation packages and any entries that couldn't be mapped were discarded. The analyses were carried out separately for each cell line on the CTR versus TIS comparison. Each consensus DAS gene was then assessed for differential expression (as per the criteria in Section 4.3) and the degree of overlap between the transcriptional and alternative splicing changes was calculated as both the absolute number and the proportion of DAS genes that were also differentially expressed.

4.11 Cross-cell-line concordance of high-confidence events

High-confidence events were those supported by both rMATS and LeafCutter in the CTR vs TIS comparison. Consensus events identified in one cell line (the reference) were matched in the other cell line by exact genomic coordinates (chromosome, strand, and splice junction coordinates) and event type. Directional concordance was defined as the proportion of the shared events that had the same sign of ΔPSI. The degree of quantitative agreement in the splicing changes between cell lines was assessed by the Pearson (r) and Spearman (ρ) correlation coefficients of the ΔPSI values. To ensure an unbiased cross-comparison, the analysis was performed in both directions, with each cell line in turn as the reference. ΔPSI was also plotted against the gene-level log2 fold change to assess whether the splicing changes were coupled to, or independent of, changes in steady-state transcriptional expression.

4.12 Representative events and visualisation

Events were selected as representative on the basis of high confidence (supported by both rMATS and LeafCutter), strict directional concordance across both cell lines, and biological relevance to pre-mRNA splicing, the spliceosome (e.g., SR proteins), and cell cycle arrest. For each gene, the most significant event (lowest FDR; ties broken by |ΔPSI|) is shown, with ΔPSI and FDR from rMATS and the steady-state expression parameters (DESeq2 normalized counts, log2 fold change and adjusted p-value) from DESeq2. Heatmaps were produced using ComplexHeatmap and circlize in R, with rows not hierarchically clustered but grouped by splicing trajectory class and ordered by the magnitude of the driving ΔPSI (color scale constrained between -0.4 and +0.4). Venn diagrams were made using ggVennDiagram and multi-set intersections using ComplexHeatmap UpSet plots. Sashimi-style junction plots and coverage tracks were generated from the BAM files using custom R scripts, with read depths normalized to reads per million (RPM).

5. Results

5.1 Global Gene-Expression Dynamics Following Therapy-Induced Senescence

5.1.1 Quality control and sample clustering

The PCA and Pearson correlation matrices demonstrated strong within-condition consistency among biological replicates in both cell lines. Samples clustered by condition, with clear separation between the TIS state and both the CTR and REPOP states. PC1

accounted for 74.3% and 81.7% of the total variance in the MCF7 and T47D cell lines, respectively, and was the principal axis separating the TIS samples from the other two conditions. PC2 captured the transcriptional differences between the CTR and REPOP states, with the REPOP state being clearly distinct from both the CTR and TIS states.

5.1.2 Global differential gene-expression

Differential gene expression analysis demonstrated extensive transcriptional changes associated with therapy-induced senescence in both cell lines (Figure 4). In the CTR vs TIS comparison, MCF7 cells showed over 4,000 upregulated and around 2,000 downregulated genes, T47D cells showed approximately 3,500 up and 1500 down. This is consistent with the notion that induction of TIS is accompanied by extensive transcriptional reprogramming in both cell types.

Crucially, the analysis in this paper was carried out using a completely independent quantification and differential expression pipeline from that used in the original Bajtai et al. (2025) [15] study. The Bajtai et al. study used STAR alignment, followed by gene-level quantification with FeatureCounts and differential expression analysis with edgeR and limma. I used RSEM for quantification and DESeq2 for differential expression analysis. Despite the different pipelines employed, the analysis came to the same conclusion in terms of the overall transcriptional profile and the major biological changes. The concordance between the two approaches provides independent computational validation of the robustness of the expression changes identified, suggesting that the key findings are not specific to any one quantification method.

The most striking changes in the MCF7 cells were an upregulation of genes including IFI6, ITGB6, PTPN21, GSN, OPN3, ABCC3 and CAV1. A number of these are involved in cell adhesion, membrane associated processes and cellular response to stress. In contrast there was a strong downregulation of genes involved in cell cycle progression and mitotic regulation (PLK1, KIFC1, KNL1, ASPM, PRR11, GTSE1 and FOXM1). This is entirely in keeping with the profound suppression of proliferative activity that is a hallmark of the TIS state [15].

In the T47D cells the most upregulated genes were IFI6, ISG15, IFI27, IFIT1, IFIT2, IFIT3, HLA-F, B2M and OASL. There's a very clear enrichment of interferon associated and immune response genes so that is indicative of a very robust immune related transcriptional response in that cell line. STC2 and PGR were the most dramatically downregulated genes.

The REPOP vs TIS comparison again shows that repopulation is associated with significant transcriptional changes. The expression trajectories are quite different between the two cell lines. In the MCF7 cells a number of genes that were strongly induced during TIS (ITGB6, OPN3, GSN, ABCC3 and LMO7) were down in the REPOP cells. Other genes (FAM234B, CSTA, WNT2B and FOSB) were relatively more highly expressed in the REPOP cells. So repopulation is associated with both the reversal of some of the TIS expression changes and the appearance of a number of new expression changes.

In the T47D cells the REPOP vs TIS comparison was very clear in terms of the reversal of the interferon associated expression signature. The genes that were strongly induced during TIS (IFI6, ISG15, IFIT2, IFIT3, HCP5, OASL and CCL5) were all dramatically down in the REPOP cells. The converse was true of STC2, PGR and EGR3 which were relatively up in the REPOP. So the very strong immune associated transcriptional response seen during TIS is pretty much switched off once cell have escaped senescence.

Overall, the differential expression analysis shows that TIS is associated with widespread transcriptional reprogramming in both MCF7 and T47D cells, although different gene sets are affected in the two cell lines. REPOP is associated with the reversal of many TIS-associated expression changes (particularly the immune signature in T47D cells), with MCF7 cells retaining a more extensive set of transcriptional differences. Taken together, this supports the transient nature of the TIS-associated transcriptional state and highlights differences in the extent and nature of transcriptional recovery.

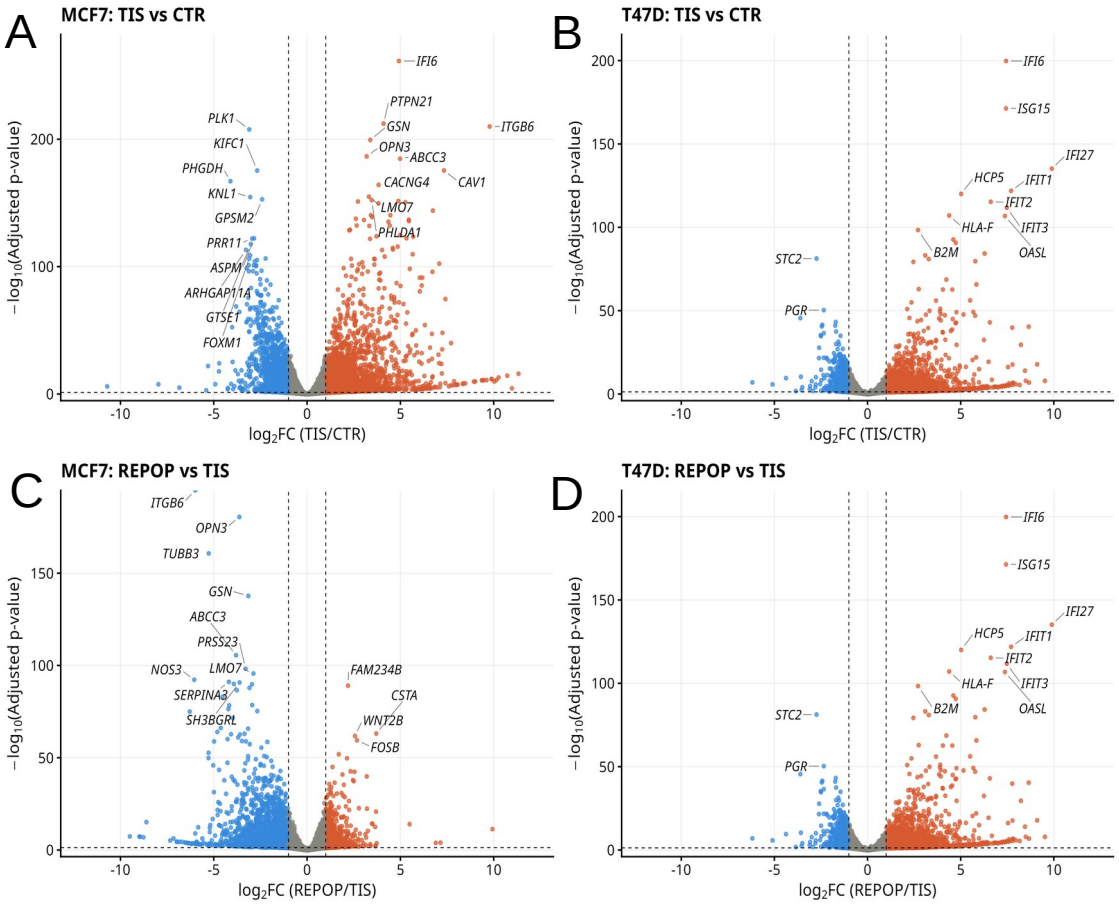


Figure 4. Differential gene expression across the control (CTR), therapy-induced senescence (TIS), and repopulation (REPOP) continuum. Volcano plots of transcriptomic changes in MCF7 (A, C) and T47D (B, D) cells comparing TIS to CTR (A, B) and REPOP to TIS (C, D).

5.2 Gene-set enrichment analysis of differential gene-expression during TIS and REPOP

To characterize the biological processes associated with the gene-expression changes that occur during therapy-induced senescence (TIS) and subsequent repopulation (REPOP), I performed GSEA on the ranked gene lists from the CTR vs TIS and REPOP vs TIS comparisons in both the MCF7 and T47D cell lines. This identified coordinated changes in the expression of genes involved in RNA processing, cell cycle regulation, DNA maintenance, inflammatory signalling and extracellular matrix organization. Of particular interest was the observation that several of the pathways were oppositely enriched between TIS induction and REPOP, indicating that there are substantial transcriptional changes associated with the transition between these two states (Figure 5).

5.2.1 Suppression of RNA processing and the spliceosome during TIS

A key finding was the downregulation of genes involved in RNA processing. In the T47D cells the KEGG Spliceosome pathway was one of the most strongly negatively enriched gene sets (only behind DNA replication in terms of strength of enrichment) with a very significant adjusted p value ($<10^{-7}$), a GeneRatio of ~0.70 and over 80 contributing genes. In the MCF7 cells the WikiPathways mRNA processing term was similarly amongst the top negatively enriched categories (adjusted $p < 10^{-7}$; GeneRatio ~0.69; over 70 contributing genes).

Together these results point to a coordinated downregulation of genes involved in spliceosome function and more general mRNA processing during TIS. This provides an important transcriptomic context for the subsequent alternative splicing analysis, in that senescence induction is associated with changes in both individual splicing events and the expression of genes encoding the components of the RNA processing machinery.

5.2.2 Downregulation of DNA replication and cell-cycle pathways

Genes involved in DNA replication and the cell cycle were also negatively enriched (again consistent with reduced expression of genes associated with cell proliferation). Further negatively enriched pathways were Base excision repair, the Fanconi anemia pathway and Homologous recombination all of which are DNA maintenance/repair pathways. All of this is again consistent with the reduced proliferative state and altered DNA maintenance profile expected in senescence.

5.2.3 Enrichment of inflammatory signalling and extracellular organization

In contrast to the negatively enriched pathways, the positively enriched gene sets were Cytokine–cytokine receptor interaction and Overview of proinflammatory and profibrotic mediators. Pathways relating to cell–cell interactions and ECM organization (focal adhesion, integrin signalling and ECM–receptor interaction) were also amongst the positively enriched.

These results point to an increased representation of genes involved in inflammatory signalling and cell-matrix interactions in the TIS expression profile. This is consistent with cellular remodelling and the secretory response associated with senescence [37].

5.2.4 Transcriptional reactivation of RNA processing and proliferative pathways during REPOP

The REPOP vs TIS comparison showed the exact opposite pattern of enrichment for a number of the pathways that were suppressed during TIS. Spliceosome function, RNA processing, ribosomal activity and DNA replication gene sets all showed positive enrichment during repopulation. So the transition into the REPOP state is accompanied by increased expression of genes involved in RNA processing, protein synthesis and DNA replication.

The positive enrichment of spliceosome associated pathways is of particular relevance in the context of the alternative splicing analysis. TIS was defined by the coordinated downregulation of genes involved in spliceosome function and mRNA processing, whereas in REPOP these same pathways were positively enriched. The positive enrichment of ribosomal pathways points to a more general restoration of biosynthetic activity and the reactivation of DNA replication associated genes is in keeping with the transition to a proliferative state.

Together, these data point to repopulation involving more than simply the reversal of the individual senescence-associated expression changes, with a much broader shift in transcription towards RNA processing, protein synthesis and cell proliferation.

5.2.5 Overall interpretation

Overall the GSEA results show extensive and dynamic transcriptional reprogramming both during TIS and in the subsequent REPOP phase. TIS is defined by the downregulation of genes involved in RNA processing, spliceosome function, DNA replication, cell cycle progression and DNA repair and by the upregulation of genes involved in inflammatory signalling, cell adhesion and extracellular matrix organization. During REPOP a number of the previously suppressed programmes (particularly spliceosome function, RNA processing, ribosomal activity and DNA replication) show the opposite enrichment pattern, as would be expected with transcriptional reactivation as the system moves into a proliferative state.

Importantly, despite the use of a different quantification and differential expression pipeline (RSEM and DESeq2) than that of Bajtai et al. (2025) [15], the present analysis recovered the same major functional patterns. The concordance of results at both the gene expression and pathway level further supports the robustness of the transcriptional changes observed. The opposing enrichment of RNA processing and spliceosome-associated genes during TIS and REPOP also provides a very useful biological context for the subsequent alternative splicing analysis. Differential gene-expression and alternative splicing are two distinct facets of transcriptomic regulation, and analyzing both is required to fully characterize the molecular changes that occur during senescence induction and subsequent repopulation.

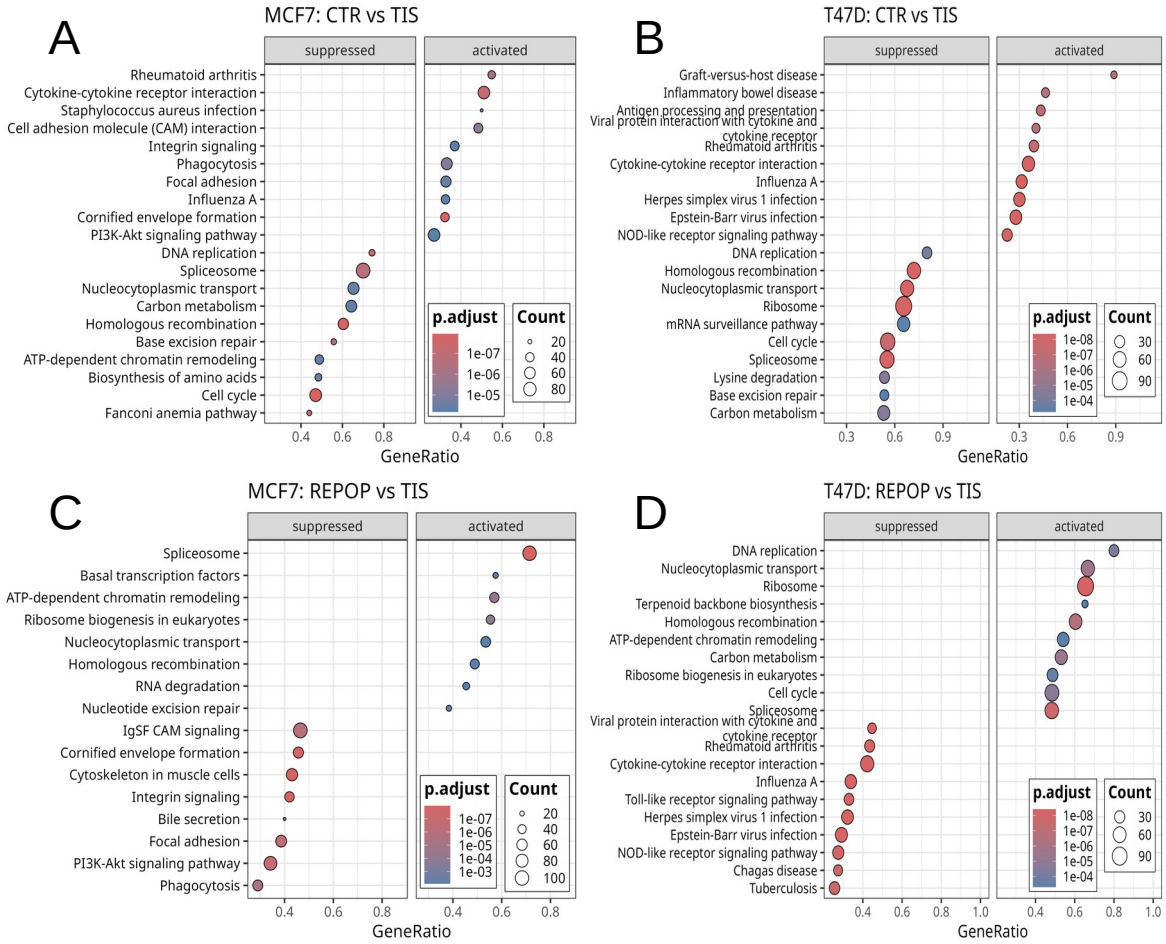


Figure 5. Pathway enrichment analysis across the control (CTR), therapy-induced senescence (TIS), and repopulation (REPOP) continuum. Dot plots displaying activated and suppressed pathways from gene set enrichment analysis in MCF7 (A, C) and T47D (B, D) cells during senescence induction (CTR vs TIS) and repopulation (REPOP vs TIS). Dot size corresponds to gene count, and color indicates the adjusted p-value.

5.3.1 Global distribution of alternative splicing changes across event types

Before examining individual AS events, I first looked at the global distribution of splicing changes to see whether the splicing changes seen during TIS could simply reflect a general failure of the splicing machinery. To that end, I looked at the distribution of ΔPSI values between TIS and CTR for all five of the canonical types of AS events: alternative 3' splice site (A3SS), alternative 5' splice site (A5SS), mutually exclusive exons (MXE), retained introns (RI) and skipped exons (SE).

As shown in (Figure 6)**,** the ΔPSI distributions remained very symmetric and centred around zero for all event types in both MCF7 and T47D cells. No obvious global shift occurred in either direction. Crucially, retained introns showed no systematic positive shift, which would otherwise suggest a general increase in intron retention due to reduced splicing efficiency.

These observations would suggest that TIS does not cause a generalised disruption of alternative splicing or a wholesale shift in splice site usage. The large number of significant splicing changes identified in the subsequent analyses occurs against a background of a very stable global splicing distribution. That supports the interpretation that TIS is associated with a selective alteration in splicing regulation rather than a non-specific collapse of the splicing machinery. That is an important context in which to view the subsequent analysis of individual splicing events and their functional consequences.

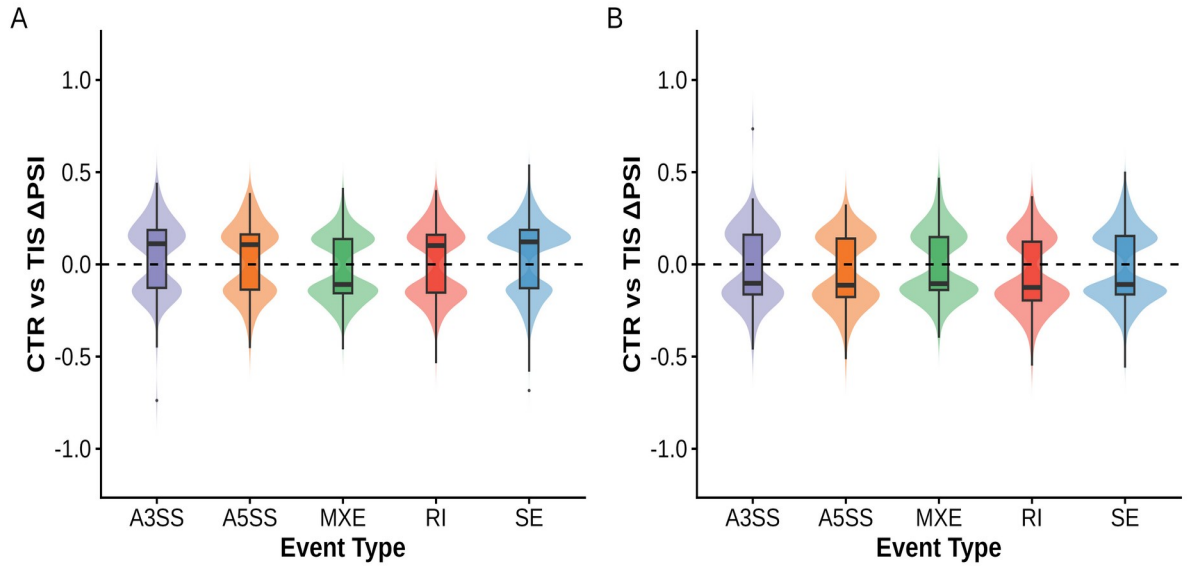


Figure 6. Distribution of splicing inclusion changes across alternative splicing event types. Violin and box plots showing the change in percent spliced in (ΔPSI) distributions between the control (CTR) and therapy-induced senescent (TIS) states across the five canonical alternative splicing classes: alternative 3' splice site (A3SS), alternative 5' splice site (A5SS), mutually exclusive exons (MXE), retained introns (RI), and skipped exons (SE) in (A) MCF7 and (B) T47D cell lines. Box plots show the median and interquartile range (IQR).

5.3.2 Cross-cell-line concordance of differential alternative splicing

Comparison of the significant rMATS events in MCF7 and T47D showed very little overlap at the event level. Of the 84,506 testable events in the common universe, 4,802 were significant in MCF7 and 2,545 in T47D, with 813 events significant in both cell lines. So, 3,989 events were specific to MCF7 and 1,732 to T47D. The corrected Jaccard index was 12.44%, indicating very little overlap between the significant event sets, even after accounting for differences in event detectability between the two cell lines (Figure 7B).

Skipped exon (SE) events were by far the most common event category in both cell lines, making up 55.1% of the significant events in MCF7 and 61.2% in T47D (Figure 7A). The relative proportions of the other event types differed substantially between the two models. In MCF7, RI and MXE events accounted for 16.5% and 15.6% of the significant events, respectively, compared to 11.5% and 12.7% in T47D. The reverse was true for A3SS

and A5SS events, which were more abundant in T47D. Despite differences in the relative proportions of event types, MCF7 had more significant events in all five event categories.

LeafCutter also performed a complementary analysis of differential intron usage. Of the 15,333 genes in the common universe that were amenable to testing, 2,948 were significant in MCF7 and 3,014 in T47D, with 1,220 in common between the two cell lines. That gave 1,728 MCF7-specific and 1,794 T47D-specific significant genes. The corrected Jaccard index was 25.73%, indicating a higher degree of overlap between the two cell lines than expected from the rMATS analysis.

Overall, both approaches showed substantial cell line-specific differences in alternative splicing, with LeafCutter showing more overlap. The methods also differed in their detection patterns, with rMATS finding more significant events in MCF7 and LeafCutter finding a slightly higher number of significant genes in T47D.

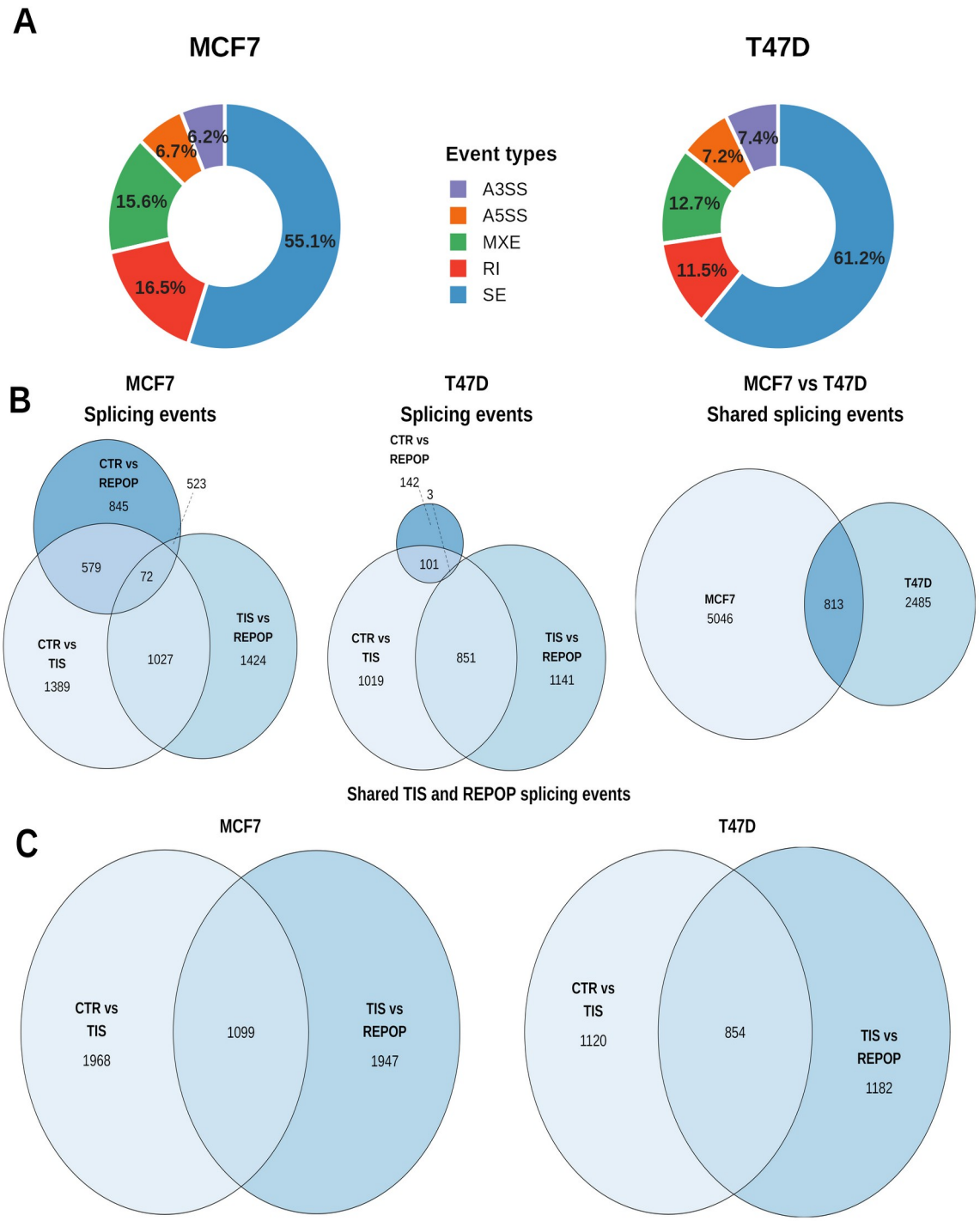


Figure 7. Distribution and overlap of alternative splicing events in MCF7 and T47D cells. (A) Proportions of canonical alternative splicing event types: alternative 3' splice site (A3SS), alternative 5' splice site (A5SS), mutually exclusive exons (MXE), retained introns (RI), and skipped exons (SE) detected in each cell line. (B) Venn diagrams showing intersections of significant splicing events across treatment comparisons: control versus therapy-induced senescence (CTR vs TIS), TIS versus repopulation (TIS vs REPOP), and CTR versus REPOP, as well as the overlap of shared events between MCF7 and T47D cell lines. (C) Shared splicing events between the CTR vs TIS and TIS vs REPOP transitions in each cell line.

5.3.3 Dynamic trajectories of alternative splicing during TIS and repopulation

Heatmap analysis of the alternative splicing changes across the CTR, TIS, and REPOP states identified six distinct trajectory patterns in both MCF7 and T47D cells: persistent, progressive, partial, transient, reversed, and REPOP-specific (Figure 8A, B). These categories describe the nature of splicing changes as they progress through the two transitions, distinguishing those maintained after TIS from those attenuated, reversed, or newly acquired during the REPOP phase. Of the significant events identified during TIS, 2,699 in MCF7 and 1,786 in T47D fell into the five TIS-dependent trajectory categories. The REPOP-specific events were analyzed separately.

Persistent changes. Persistent events showed a clear splicing shift during the CTR to TIS transition, with little further change during the repopulation phase. The changes seen during TIS were therefore largely maintained in REPOP. This group made up 26.9% (726 events) of the classified trajectories in MCF7 and 14.4% (257 events) in T47D. Of note, 62.8% of the MCF7 events and 31.1% of the T47D events also remained statistically significant in the REPOP versus CTR comparison. Examples were SPPL2A, EIF4A2, CDH24 and ZNG1E in MCF7 and ZNF227 and LRRC23 in T47D. So a subset of the splicing changes associated with TIS are persistent and remain detectable after the cells have repopulated.

Progressive changes. Progressive events also showed further changes in the same direction during repopulation, reinforcing the initial TIS-associated splicing shift. This was the rarest TIS-dependent trajectory (3.5% (94 events) in MCF7 and 0.4% (8 events) in T47D). They were nevertheless statistically significant in the REPOP vs CTR comparison in most cases (96.8% in MCF7), and examples included PLEKHG3 and PSMB3 in T47D. So again, while they are very rare, progressive trajectories show that some splicing changes continue to progress rather than plateau or revert to the original state during repopulation.

Partial recovery. Partial trajectories were defined by an initial splicing shift during TIS which was then reversed in the opposite direction during repopulation without complete restoration of the original splicing pattern. This was the most common category by far (24.7% of MCF7 and 32.6% of T47D trajectories). In most cases, the initial changes were not ultimately significant in the REPOP versus CTR comparison (88.5% of MCF7 and 97.1% of T47D). Examples included ANKRD10 and EP400 in MCF7 and ZFAND1 in T47D. The upshot is that many splicing changes associated with TIS are attenuated during the repopulation phase, even if the trajectories do not fully recover the original splicing pattern.

Transient changes. Transient events showed a clear splicing shift during TIS, followed by the opposite shift during repopulation, which effectively brought the splicing pattern back to baseline. This was the predominant trend in both cell lines (38.3% of events in MCF7 (1,034 events) and 51.1% in T47D (912 events)). Consistent with this, 99.6% of MCF7 events and 100% of T47D events in this category were no longer significant in the REPOP versus CTR comparison. Examples include PFKM and DUXAP10 in MCF7 and THOC5 and POMT1 in T47D. The sheer number of events in this category suggests that a large proportion

of the splicing changes associated with TIS are not persistent after the period of proliferative recovery.

Reversed changes. In contrast to the transient events, the reversed trajectories showed an opposing change during repopulation which was greater in magnitude than the original TIS-associated shift, resulting in a final splicing pattern on the other side of the control. This was a fairly rare category, accounting for 6.6% of MCF7 and 1.5% of T47D trajectories. The vast majority of these events (87.1% in MCF7) did not remain statistically significant in the REPOP vs CTR comparison. Examples were LRP8 and SNHG21 in MCF7 and PDPR2P and SH2D3A in T47D. The low frequency of these events suggests that large overshoots beyond the original splicing state are uncommon.

REPOP-specific changes. In addition to the five TIS-dependent trajectories, another group of events showed little change during the CTR to TIS transition but did show significant splicing changes during repopulation. The number of these events varied considerably between the cell lines: 1,262 in MCF7 vs 122 in T47D. Examples were ENSG0000029083 and SNHG32 in MCF7 and MAP4K3-DT and ZNF37A in T47D. These events show that repopulation is not simply a process of reversal or maintenance of senescence-associated changes, but is also associated with a suite of splicing changes specific to the repopulation phase.

Trajectory robustness. Some trajectory classes (in particular the reversed events) are very sensitive to the magnitude of the initial splicing change, as they depend on the relative size of the change from one time point to the next. To test this, I re-calculated the trajectory distributions using increasingly strict thresholds for the initial splicing shift. Moving from an inclusion threshold of $|\Delta PSI| > 0.1$ to $|\Delta PSI| > 0.2$ yielded stable, or even increased, proportions of the major trajectory classes. For instance, in MCF7, transient events increased from 38.3% to 42.5%, and reversed events decreased from 6.6% to 3.7%. The median initial ΔPSI of the reversed events was around 0.14, which is consistent with the idea that this class is particularly sensitive to small initial changes. The fact that the major trajectory classes are so well maintained under these more stringent thresholds suggests that the overall distribution is not simply a reflection of the events with the very smallest initial splicing changes.

Taken together, the trajectory distributions show quite distinct splicing dynamics in the two cell lines. MCF7 has a fairly broad distribution across the trajectory categories with a very large number of REPOP-specific events. In T47D, partial and transient trajectories dominate (together accounting for over 83% of TIS-dependent events), while progressive and reversed trajectories are relatively rare. The degree and direction of splicing changes during repopulation are therefore highly cell line dependent.

Overall, the trajectory analysis shows that alternative splicing changes heterogeneously from TIS through repopulation. Many senescence-associated changes are attenuated or reversed, some are lost, and others progress or are replaced by entirely new changes. The repopulating state is therefore not simply a reversion to the control-associated splicing profile but is a distinct state with its own pattern of post-transcriptional regulation.

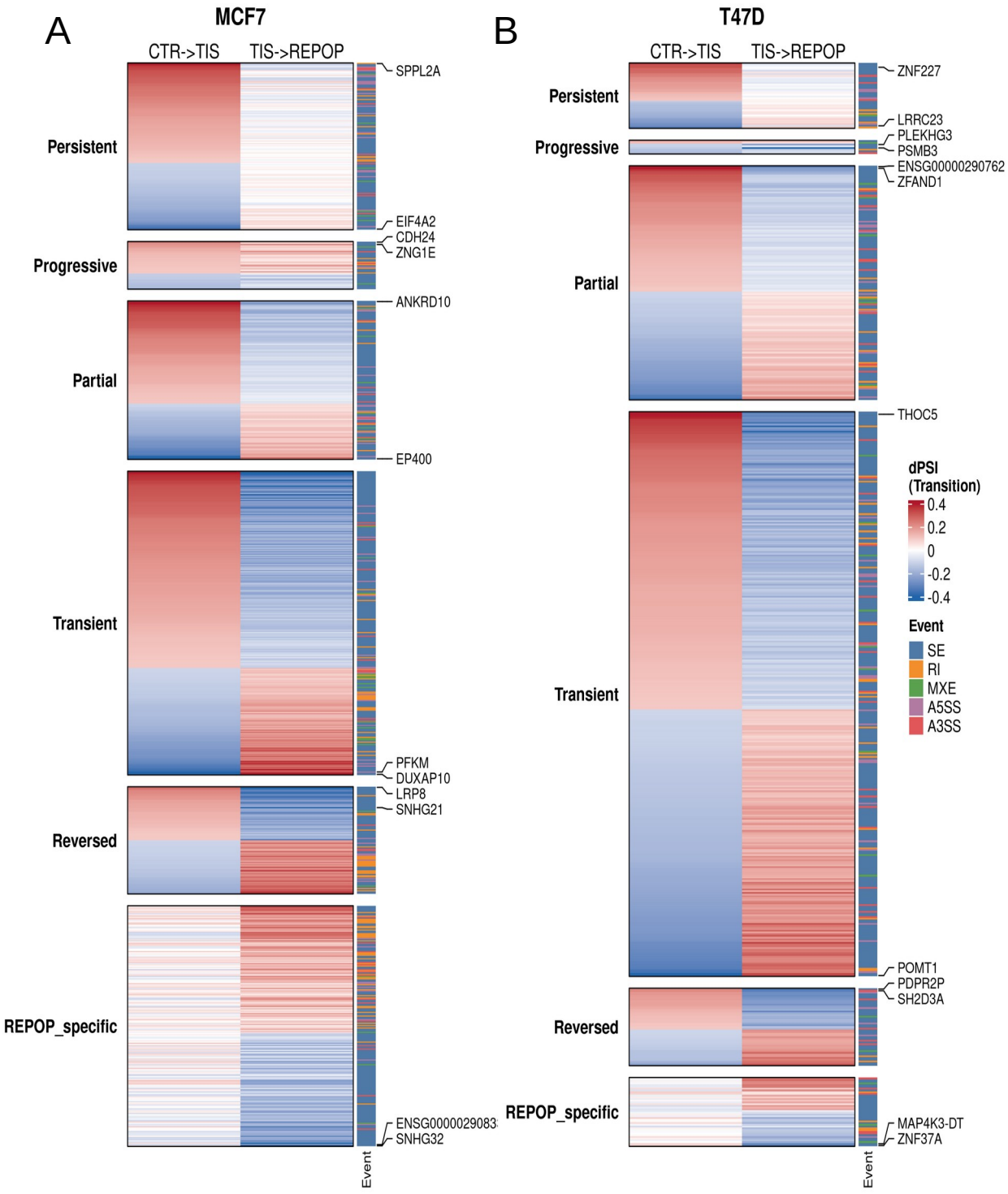


Figure 8. Clustering of alternative splicing dynamics across the control (CTR), therapy-induced senescence (TIS), and repopulation (REPOP) continuum. Heatmaps showing change in percent spliced in (ΔPSI) changes across transitions (CTR -> TIS and TIS -> REPOP) grouped by trajectory patterns (persistent, progressive, partial, transient, reversed, and REPOP-specific) in MCF7 (A) and T47D (B) cells, with canonical alternative splicing event types: alternative 3' splice site (A3SS), alternative 5' splice site (A5SS), mutually exclusive exons (MXE), retained introns (RI), and skipped exons (SE) annotated on the right.

5.4 Integration of rMATS and LeafCutter algorithms

5.4.1 Method-dependent detection and recurrence of alternative-splicing-associated genes

The rMATS vs LeafCutter comparison showed a very large difference in the number of genes called as having significant alternative splicing events. LeafCutter in all cases called significantly more genes as having significant AS events than rMATS in both cell lines (the size of the difference varied between the conditions). The difference was particularly striking in the CTR–REPOP comparison in the T47D cells. This is likely to be due to the different ways the two methods work: rMATS looks at pre-defined classes of AS events whereas LeafCutter looks at differential intron usage and does not require you to pre-define the AS events. They therefore capture different aspects of the transcript processing and the results should be viewed as complementary rather than as being directly comparable.

The gene level overlap analysis also revealed quite different patterns of recurrence across the various comparisons. Leafcutter in particular showed a very high degree of overlap between the CTR–TIS and TIS–REPOP contrasts (especially in the T47D cells). This suggests that a large number of genes were associated with significant splicing changes in both of those transitions. That said, the overlap does not necessarily mean that the same splicing events were altered in both cases, nor does it mean that the changes were in the same direction. It could be that different introns or different splicing events within the same gene were responsible for the observed overlap.

The rather small intersection of genes identified across all six method-comparison sets (65 in MCF7 and 38 in T47D) further underlines the method and comparison dependent nature of the significant splicing associated findings (Figure 9). This reinforces the need to use a range of complementary methods and to look at individual events rather than just gene level significance overlap.

Overall these data suggest that the identification of alternative splicing associated genes is dependent on both the method of analysis and the cell transition being studied. The frequency with which genes recur across the various comparisons gives a broad picture of potentially important regulatory targets, while the direction and extent of the individual splicing changes will need to be investigated in isolation.

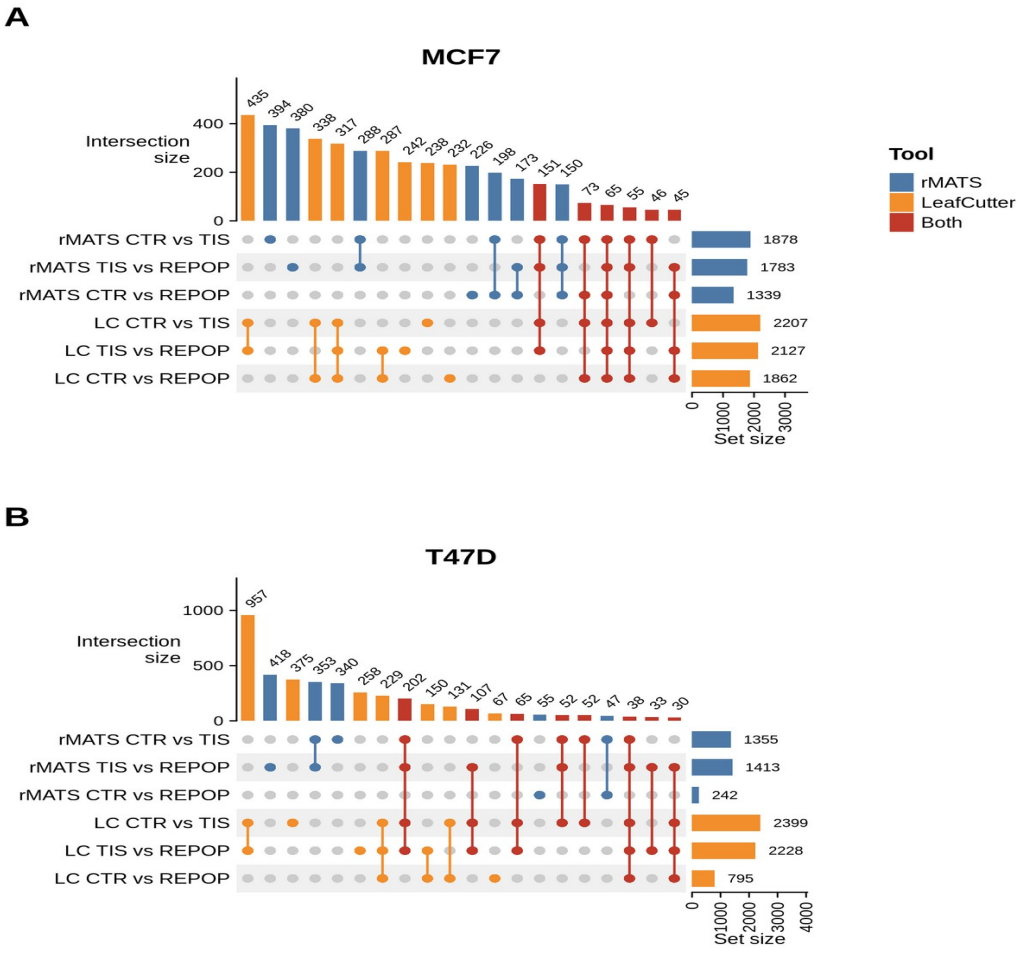


Figure 9. Overlap of differentially spliced genes across conditions and detection methods. UpSet plots illustrating intersections of significant splicing-affected genes across pairwise treatment comparisons: control versus therapy-induced senescence (CTR vs. TIS), TIS versus repopulation (TIS vs. REPOP), and CTR versus REPOP identified by rMATS and LeafCutter in (A) MCF7 and (B) T47D cells. Horizontal bars on the right indicate the total number of significant genes detected per tool and comparison.

5.4.2 Directional concordance between rMATS and LeafCutter

To assess the agreement between rMATS and LeafCutter beyond the sheer overlap of significant findings, I evaluated the directional concordance of the splicing changes identified by the two methods across both cell lines. Crucially, to avoid the statistical tautology of comparing only dual-filtered sets, this analysis was performed on an unfiltered background of 6,896 shared events. In this approach, no significance or effect size filtering was applied to the LeafCutter output prior to matching with the rMATS events. Following the harmonization of effect directions, a remarkable 98% of these unfiltered matched results exhibited concordant directions of change in the pooled analysis (Figure 10).

This exceptionally high level of overall concordance demonstrates that the two independent algorithms strongly agree on the underlying biological trajectory—whether an event is increasingly included or excluded—even outside the strict bounds of significance thresholds. While the intersection metrics highlighted differences in the sheer number and identity of statistically significant events flagged by each tool, the unfiltered directional analysis confirms that when both tools detect a quantifiable signal, their biological interpretation is almost perfectly aligned.

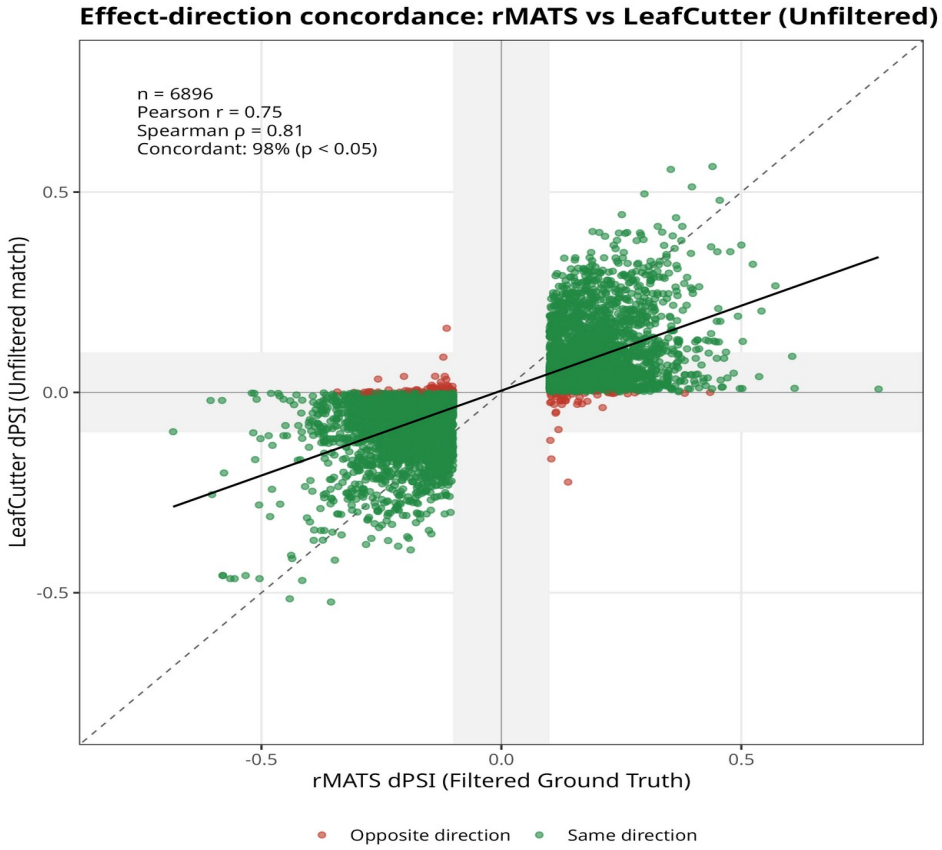


Figure 10. Effect-direction concordance between rMATS and LeafCutter. Scatter plot comparing splicing inclusion changes of coordinate-matched skipped exon (SE) events (n = 6896) without LeafCutter pre-filtering. Splicing changes show 98% directional agreement (green: same direction, red: opposite direction), with strong correlation across methods (Pearson r = 0.75, Spearman rho = 0.81, p < 0.05).

5.5 Functional enrichment of genes with shared alternative-splicing changes

To characterize the biological processes associated with genes that showed shared alternative splicing changes during therapy-induced senescence (TIS), over-representation analysis (ORA) was carried out. The most significantly enriched terms across the

CANONICAL_ALL, GO_BP, and HALLMARK databases were related to RNA processing and splicing, with additional enrichment in cell cycle regulation, mitotic organization, and genome maintenance (Figure 11).

5.5.1 RNA processing and the splicing machinery

RNA metabolism was by far the most prominent functional category. In the CANONICAL_ALL collection, the top enriched terms were 'Processing of capped intron-containing pre-mRNA', 'mRNA splicing', 'mRNA 3' end processing' and 'mRNA polyadenylation'. This was borne out in the GO_BP results, where 'RNA splicing' was by far the most significant term, with 'RNA splicing via transesterification reactions' and 'regulation of RNA splicing' also in the top tier. The consistent enrichment of these terms across databases suggests that genes with shared TIS-associated splicing changes are associated with the machinery of pre-mRNA processing and maturation. The observed splicing changes may therefore reflect not only individual target transcripts but the splicing regulatory network itself.

5.5.2 Cell-cycle regulation and mitotic organization

Beyond RNA metabolism, the enriched categories pointed to alterations in cell cycle regulation and mitotic organization. The HALLMARK analysis identified significant enrichment of the 'G2M checkpoint' gene set, and the GO_BP terms included 'spindle organization', 'microtubule cytoskeleton organization involved in mitosis', and 'mitotic spindle organization'. Thus, the shared splicing changes also affect genes associated with cell cycle control and mitotic spindle organization, processes intimately linked to the proliferative arrest that is the hallmark of TIS.

5.5.3 DNA repair and cellular homeostasis

Additional enriched biological processes related to genome maintenance and cellular organization. 'DNA recombination' and 'double-strand break repair' were significantly enriched within the GO_BP categories, pointing to genes involved in DNA damage repair and genome stability. 'Macroautophagy' and 'membraneless organelle assembly' were also significantly enriched, suggesting links to intracellular homeostasis and cellular structure organization.

Overall, ORA showed that genes with shared alternative splicing changes during TIS cluster into interrelated processes, most notably RNA processing and splicing, followed by cell cycle regulation, mitotic organization, and DNA repair. The strong enrichment of RNA processing components, in particular, suggests that the observed splicing changes may occur

in both downstream target genes and the RNA splicing machinery itself. This provides a functional framework for the shared TIS-associated splicing program and identifies several processes as candidates for further mechanistic investigation.

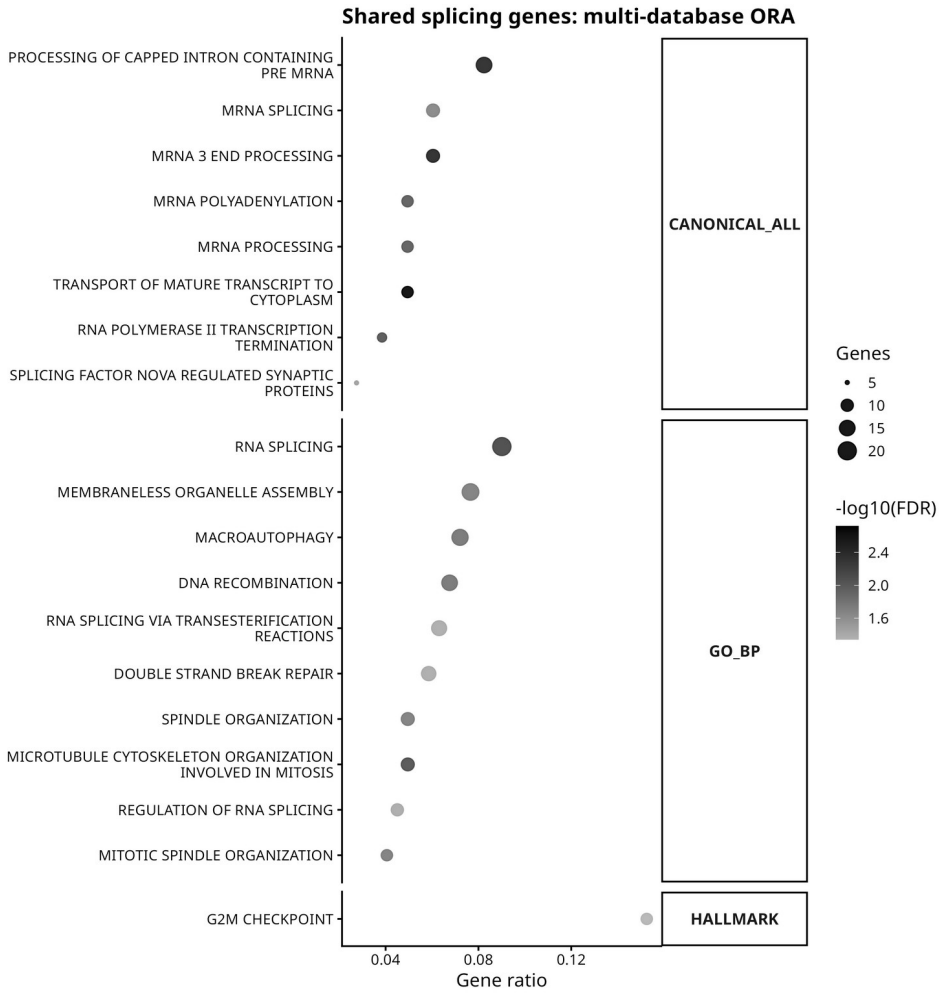


Figure 11. Multi-database over-representation analysis (ORA) of shared splicing genes during therapy-induced senescence. Top FDR-significant enriched terms across canonical pathway collections (CANONICAL_ALL), Gene Ontology (GO_BP), and Hallmark databases, highlighting prominent enrichment in RNA processing and cell-cycle pathways.

5.6 Relationship between DAS and DGE

To assess whether alterations in alternative splicing correlated with changes in total gene expression, I examined the overlap between DAS genes and DGEs across the CTR, TIS, and REPOP conditions in both MCF7 and T47D cells. I looked at the three pairwise comparisons (CTR vs TIS, TIS vs REPOP, and CTR vs REPOP). In addition to looking at the

overlap between DAS and DGE genes, I also looked at the proportion of DAS genes that were classified as DGEs and compared this to the background level of DGEs.

5.6.1 Overlap between DAS and DGE genes

In the CTR versus TIS comparison, 99 of the 674 DAS genes identified in MCF7 cells were also classified as differentially expressed (14.7% of DAS genes) (Figure 12). The remainder (575 genes, 85.3%) showed significant alternative splicing changes but did not meet the statistical significance criterion for DGE. In T47D cells, the overlap was smaller, with only 36 of the 509 DAS genes (7.1%) being classified as differentially expressed and 473 (92.9%) failing to meet the DGE significance criterion.

So, in both cell lines, the vast majority of genes that were showing differential alternative splicing going from control to therapy-induced senescence were not also genes that were differentially expressed. This was especially the case in the T47D cells, where fewer than 1 in 10 of the DAS genes were also differentially expressed.

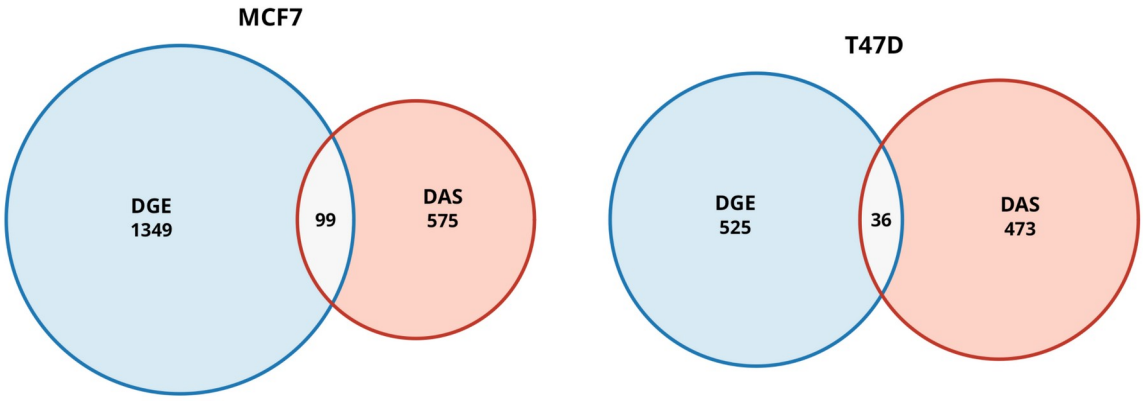


Figure 12. Overlap between differentially expressed (DGE) and alternative splicing (DAS) genes during therapy-induced senescence. Venn diagrams illustrate the limited intersection between transcriptional changes and splicing alterations in MCF7 and T47D cell lines.

5.6.2 Cross-method and cross-cell-line concordance of AS

To further assess the robustness and biological conservation of the identified splicing alterations, high-confidence events independently supported by both rMATS and LeafCutter across the two cell lines were compared. This allowed to determine which splicing events were genuinely robust and which were likely to be cell line-specific, and to see if the same individual events behaved in a similar way in the two different cell lines.

The link between differential gene-expression and alternative splicing is that significant splicing changes are not confined to genes showing dramatic expression changes. You see both positive and negative ΔPSI values across a wide range of log2 fold changes, which tells you that changes in relative exon usage are not necessarily mirrored by changes in total transcript abundance. This reinforces the idea that alternative splicing is an important additional layer of regulation, not simply a downstream effect of differential gene expression (Figure 13A, B and Figure 14A, B).

The cross-cell-line comparison showed a fairly good degree of directional conservation. In the MCF7-referenced analysis, 639 events matched, and 68.2% showed the same direction of splicing change in T47D (Figure 13C). The analysis using T47D as the reference yielded 499 matched events, and 69.7% showed the same direction of change in MCF7 (Figure 14C). Quantitative agreement was less impressive, with Pearson correlation coefficients of 0.41 and 0.35 and Spearman correlation coefficients of 0.43 and 0.44, respectively.

Together, these data suggest that the TIS-associated alternative splicing response has both shared and cell-line-dependent elements. The direction of agreement suggests that some of the splicing changes may be common features of the response to therapy-induced senescence; the moderate correlation levels indicate that the degree of regulation is cell-line dependent. The fact that ~30% of events show discordant direction also underscores the extent of cell-line-specific regulation. The analysis is restricted to events independently supported by both methods, so the observed concordance applies only to this high-confidence subset rather than the full alternative splicing landscape.

Overall, the cross-cell-line comparison further supports the reproducibility of a subset of TIS-associated splicing changes and shows that their regulation is not fully conserved between MCF7 and T47D.

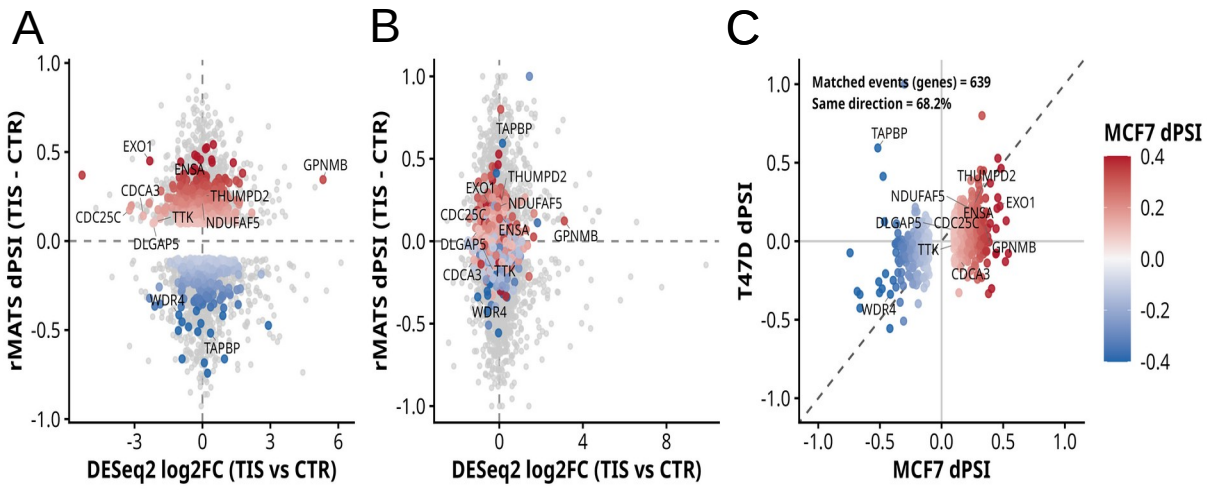


Figure 13. Relationship between differential splicing and gene expression, and cross-cell-line concordance. Scatter plots show rMATS change in percent spliced in (ΔPSI) versus DESeq2 log2 fold change (log2FC) comparing therapy-induced senescence (TIS) to control (CTR) in MCF7 (A) and T47D (B) cells. (C) Correlation of ΔPSI values for high-confidence events between cell lines.

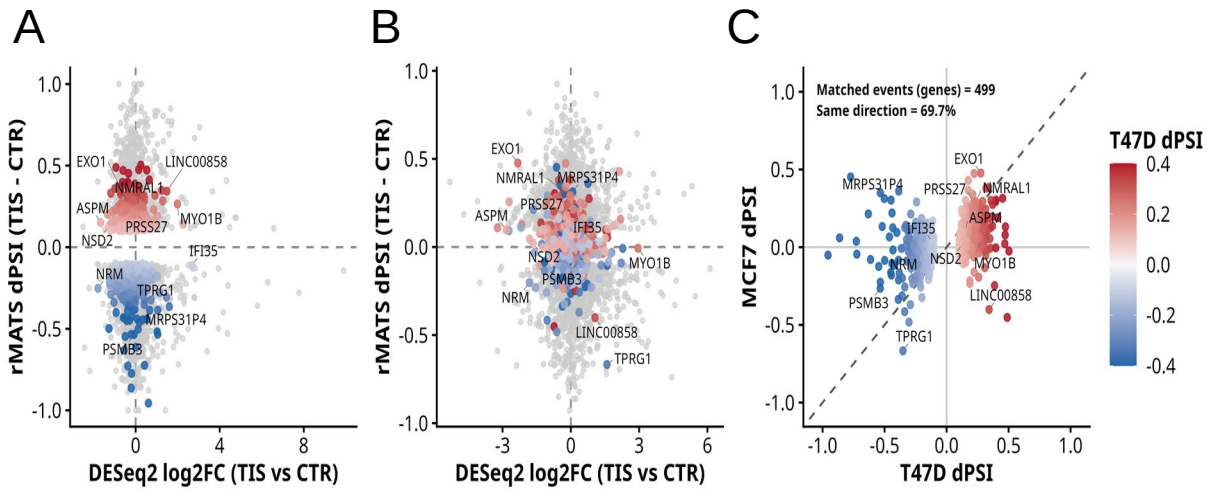


Figure 14. Relationship between differential splicing and gene expression, and cross-cell-line concordance. Scatter plots show rMATS change in percent spliced in (ΔPSI) versus DESeq2 log2 fold change (log2FC) comparing therapy-induced senescence (TIS) to control (CTR) in T47D (A) and MCF7 (B) cells. (C) Correlation of ΔPSI values for high-confidence events between cell lines.

5.6.3 Gene-Specific Splicing Changes Independent of Gene-Expression

The PSMB3 locus in T47D cells provides a clear example of substantial alternative splicing changes occurring independently of overall gene expression. Total PSMB3 expression remained essentially unchanged across the CTR, TIS and REPOP states, whereas its local splicing pattern underwent a pronounced shift. In the CTR state, exon inclusion was predominant. During TIS, exon skipping significantly increased (ΔPSI = −0.124; FDR = 0.0062). This shift became even more pronounced in REPOP, where the expression-weighted splice composition revealed a further reduction in exon inclusion relative to CTR (ΔPSI = −0.603; FDR = 5.9×10^{-7}). Thus, rather than reverting towards the control splicing pattern, PSMB3 exhibited a progressive shift towards exon skipping during repopulation, despite largely unchanged total gene expression (Figure 15).

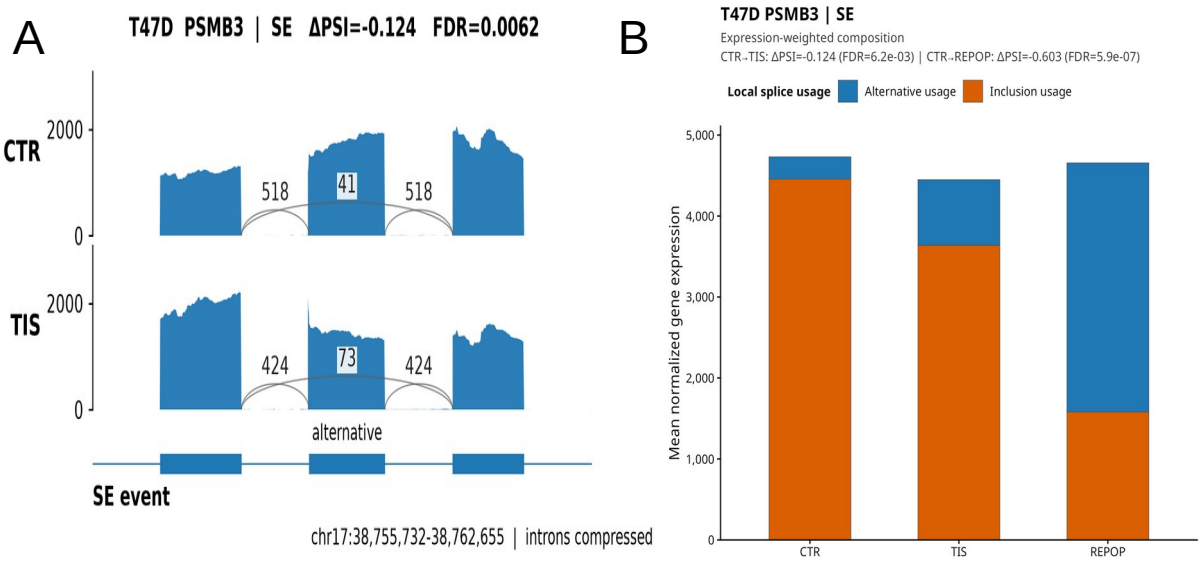


Figure 15. Representative Sashimi plot (A) and expression-weighted splice composition (B) of the PSMB3 gene in T47D cells. The A panels show read coverage and splice junctions for control (CTR) and therapy-induced senescence (TIS) states across an alternative skipped exon (SE) event, while the B bar plot displays the corresponding shift in local splice usage across the control (CTR), therapy-induced senescence (TIS), and repopulation (REPOP) continuum.

In MCF7 cells, THUMPD2 exhibited a significant skipped exon (SE) event during TIS compared to the CTR state (ΔPSI = +0.256; FDR = 1.3 x 10^{-6}). While overall gene expression was relatively stable between CTR and TIS, there was a marked change in transcript composition with an increased proportion of the exon inclusion isoform in the TIS state. This splicing change was only partially reversed upon REPOP, with ΔPSI still elevated relative to CTR (+0.198) and a modest increase in total THUMPD2 expression (Figure 16).

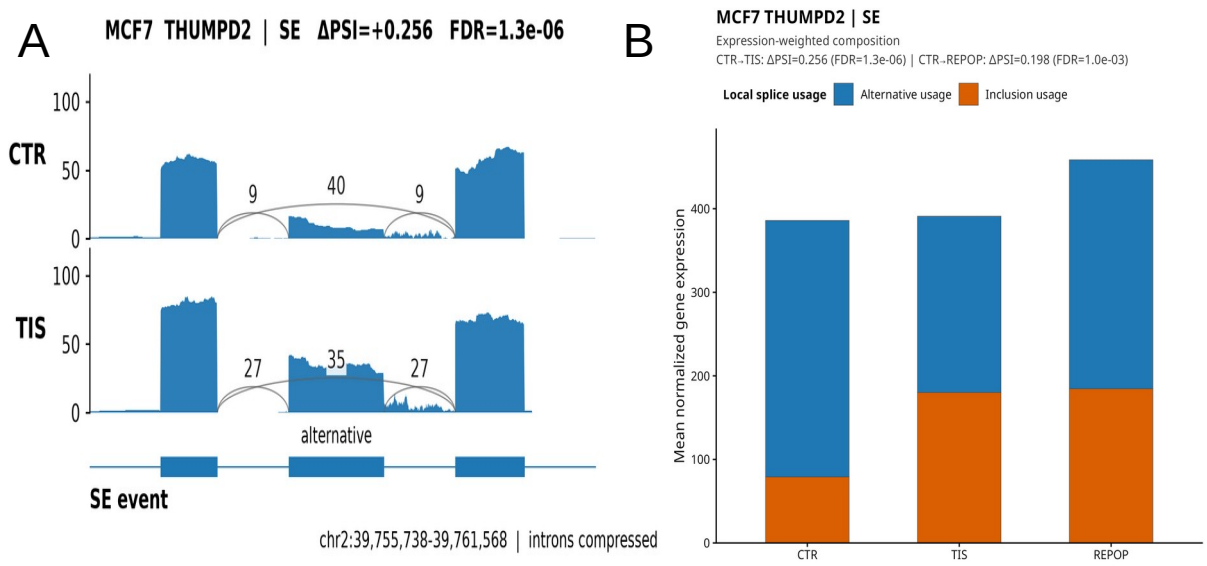


Figure 16. Representative Sashimi plot (A) and expression-weighted splice composition (B) of the THUMPD2 gene in MCF7 cells. The A panels show read coverage and splice junctions for control (CTR) and therapy-induced senescence (TIS) states across an alternative skipped exon (SE) event, while the B bar plot displays the corresponding shift in local splice usage across the control (CTR), therapy-induced senescence (TIS), and repopulation (REPOP) continuum.

5.7 Reversibility analysis

To compare the extent and time course of splicing changes between MCF7 and T47D cells, mean absolute trajectory profiles for the common splicing events across the CTR, TIS, and REPOP states (Figure 17). For each trajectory cluster, the mean absolute change in PSI relative to control (|ΔPSI|) was calculated. Taking absolute values prevents positive and negative changes from cancelling, providing a measure of the overall magnitude of splicing change in each cluster.

A striking observation is the high similarity in the trajectory profiles of the two cell lines. The trajectories of MCF7 (blue) and T47D (red) are very similar across the major clusters, suggesting that the magnitude and time course of splicing changes are essentially the same in both cell lines.

The Transient cluster really brings this pattern into focus. Both cell lines show a marked increase in mean absolute ΔPSI during TIS (around 0.20), followed by a large drop during the repopulation phase, back to control levels. So the splicing changes that define this cluster are essentially reversed upon removal of the senescent-inducing treatment.

In contrast, the Persistent cluster shows a very different pattern. There is a similarly dramatic shift during TIS, and the mean absolute ΔPSI stays high throughout repopulation (>0.17 in both cell lines). These events remained altered relative to control splicing profiles after the cells returned to proliferative conditions, consistent with the persistence of TIS-associated splicing changes during repopulation.

The Partial and Progressive clusters also show very similar profiles between the two cell lines. In the Partial cluster, splicing changes are incompletely reduced during the repopulation phase, whereas in the Progressive cluster, the magnitude of splicing changes actually increases. In both cases, the MCF7 and T47D trajectories are very similar.

Finally, the REPOP-specific cluster is defined by minimal change during TIS and a substantial increase in mean absolute ΔPSI during repopulation (approx 0.15-0.17). The very similar profiles in both cell lines again suggest these splicing changes are primarily a feature of the transition back into the proliferative state rather than the initial senescence response.

Overall, the mean absolute trajectory profiles show a high degree of concordance between MCF7 and T47D cells in both the magnitude and temporal dynamics of the shared splicing changes. The different outcomes observed during the repopulation phase (reversal, persistence, incomplete recovery, and the appearance of new changes) are clear, but the trajectory profiles are very similar between the two cell lines.

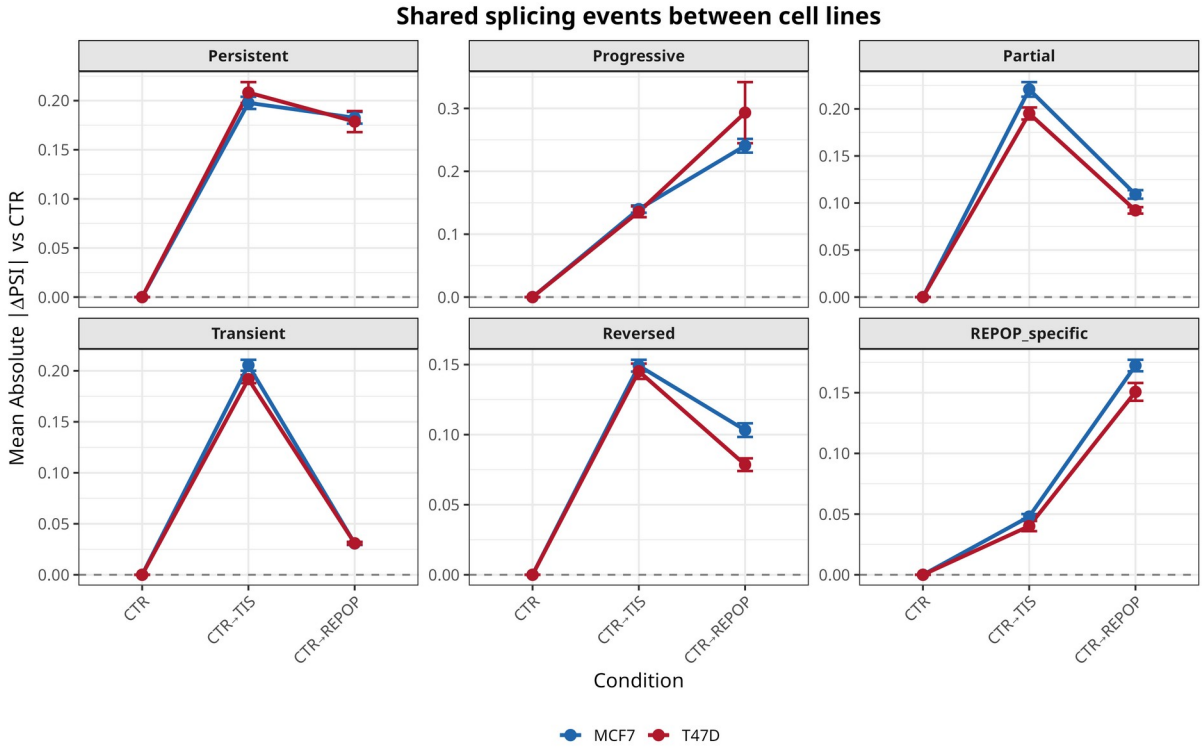


Figure 17. Trajectory profiles of shared alternative splicing events between MCF7 (blue) and T47D (red) cell lines across the control (CTR), therapy-induced senescence (TIS), and repopulation (REPOP) continuum. Line plots display the mean absolute change in inclusion levels (|ΔPSI| relative to CTR) partitioned into six trajectory classes (Persistent, Progressive, Partial, Transient, Reversed, and REPOP-specific), illustrating high concordance in magnitude and temporal dynamics. Error bars represent standard errors of the mean.

5.8 Functional context of representative alternative-splicing events during therapy-induced senescence

5.8.1 RPS24 - ribosome function and protein synthesis

RPS24 encodes ribosomal protein eS24 which is a component of the 40S small ribosomal subunit and is involved in protein synthesis. There are multiple annotated transcript variants but the functional consequences of the exon usage change will be transcript region dependent and will vary according to the effect on the RNA or protein [27].

In therapy-induced senescence (TIS) RPS24 showed a consistent increase in cassette exon inclusion in both of the breast cancer cell lines, albeit with very different magnitudes between the models. The change was dramatic in the MCF7 cells (ΔPSI = +0.524, FDR < 1 x 10^{-300}) and much more modest in the T47D cells (ΔPSI = +0.130, FDR < 1 x 10^{-300}) (Figure 18). So both models showed the same direction of change in alternative exon usage, just to differing extents.

At the gene level the expression profiles were quite different between the cell lines. In the MCF7 cells RPS24 expression was fairly constant with mean Deseq2 normalized counts going from 28,012 in the CTR to 32,030 in the TIS (log2fc = +0.19, adj p = 0.25, not significant). In the T47D cells expression went down from 45,519 in CTR to 27,124 in TIS (log2fc = −0.73, adj p = 1.4 x 10^{-7}) which was a significant decrease.

These data clearly separate the regulation of relative exon usage from changes in overall gene expression. There was increased exon inclusion in both cell lines even though the two cell lines showed very different gene level expression responses. The splicing shift therefore did not occur in parallel with a general change in expression. It is not yet known whether the affected exon has any impact on RPS24 transcript stability, coding potential ribosomal protein function (this would require annotation of the transcript and functional validation).

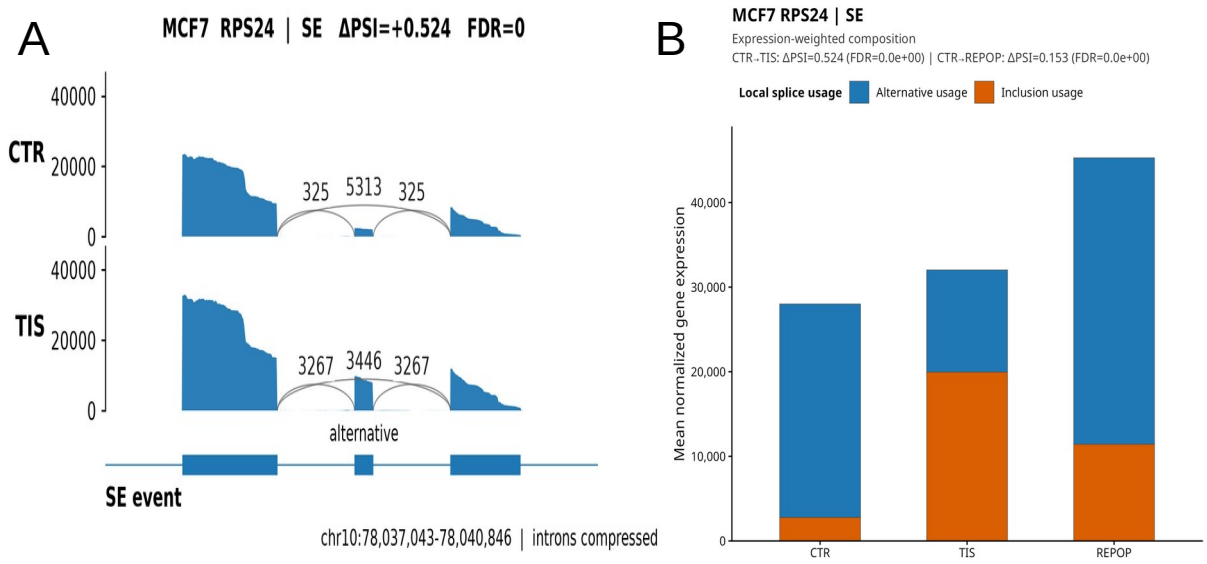


Figure 18. Representative Sashimi plot (A) and expression-weighted splice composition (B) of the RPS24 gene in MCF7 cells. The A panels show read coverage and splice junctions for control (CTR) and therapy-induced senescence (TIS) states across an alternative skipped exon (SE) event, while the B bar plot displays the corresponding shift in local splice usage across the control (CTR), therapy-induced senescence (TIS), and repopulation (REPOP) continuum.

5.8.2 ZNF207 - Cell Line-Dependent Regulation of Exon Inclusion

ZNF207 encodes a protein which is involved in mitotic spindle assembly and chromosome segregation. During TIS there was a cell line dependent pattern of alternative splicing of ZNF207 with a greater increase in cassette exon inclusion in MCF7 than in T47D.

In MCF7 cells total ZNF207 expression was almost the same, the mean Deseq2 normalized counts dropping only slightly from ~5,300 in the CTR to ~5000 in the TIS. The selected cassette exon event however showed a dramatic increase in inclusion (ΔPSI = +0.261), with the estimated inclusion fraction increasing from ~40% in the CTR to ~70% in the TIS. So there was a very significant change in the relative usage of the exon without any change in overall gene expression (Figure 19).

T47D cells had a different baseline expression level and a much more modest splicing response. ZNF207 expression was stable at around 10,000 normalized counts in both the CTR and TIS conditions. The selected event did however show a significant increase in exon inclusion (ΔPSI = +0.203; FDR = 0.0056), the inclusion fraction going from around 55% to 75%.

Together these data indicate that the TIS-associated alternative splicing of ZNF207 is in the same direction in both cell lines but differs in extent. In both cell lines there is an increase in exon inclusion despite essentially no change in gene level expression. This

demonstrates that alternative splicing can alter the transcript profile without any change in overall transcript abundance. The greater shift in MCF7 further underlines the cell line specific nature of the splicing response.

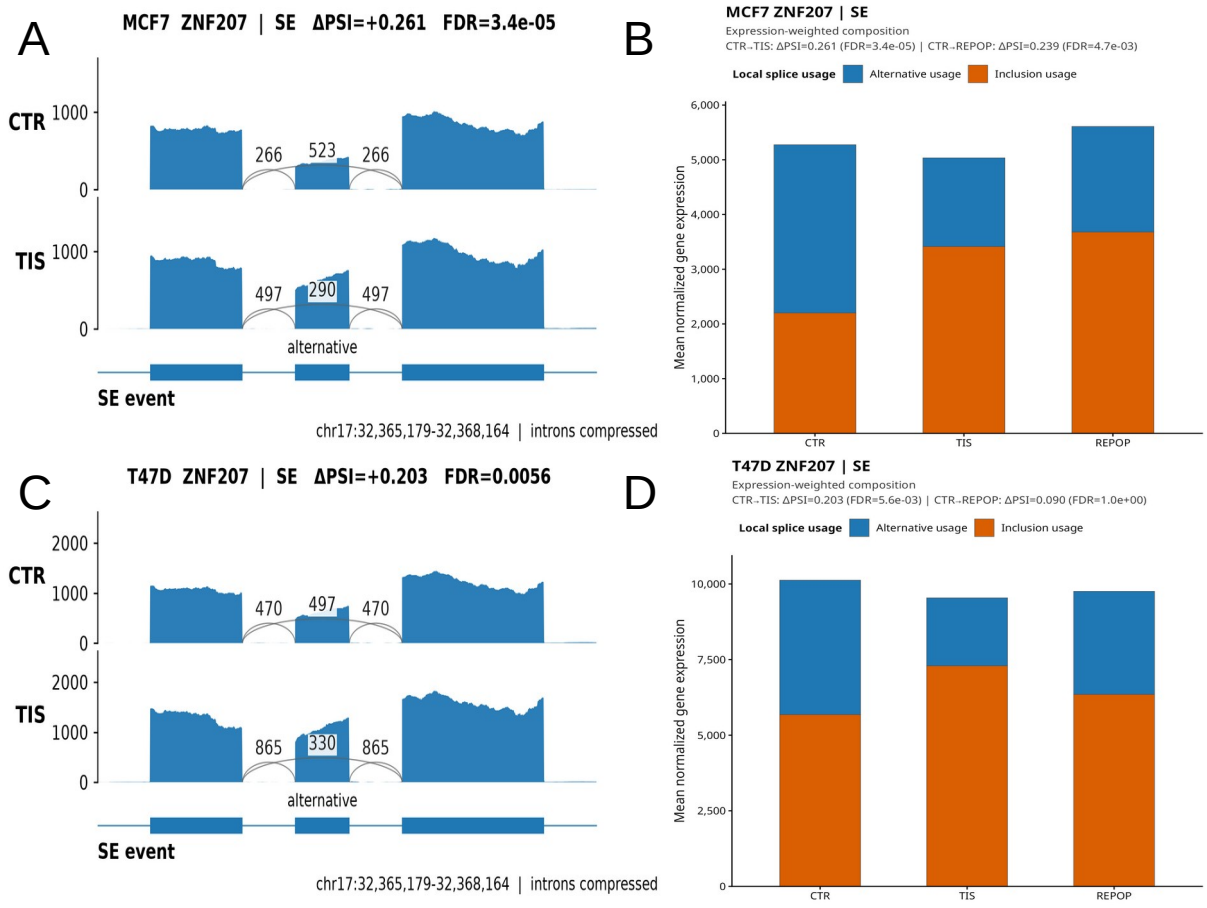


Figure 19. Representative Sashimi plots (A, C) and expression-weighted splice compositions (B, D) of the ZNF207 gene in MCF7 (A–B) and T47D (C–D) cells. The A and C panels show read coverage and splice junctions for control (CTR) and therapy-induced senescence (TIS) states across an alternative skipped exon (SE) event, while the B and D bar plots display the corresponding shifts in local splice usage across the control (CTR), therapy-induced senescence (TIS), and repopulation (REPOP) continuum.

5.8.3 SRSF1 - A Splicing Regulator Associated with Therapy-Induced Senescence

SRSF1 (which encodes a serine/arginine-rich splicing factor involved in the regulation of pre-mRNA processing) showed large changes in both gene expression and alternative splicing in both MCF7 and T47D cells. In both cell lines, SRSF1 expression was markedly reduced during TIS, although the degree of recovery during repopulation was different. In MCF7 cells, expression partially recovered during REPOP, whereas in T47D cells it returned almost completely to control (CTR) levels. Both cell lines shifted significantly toward the exon-skipping isoform during TIS (MCF7: $\Delta PSI = -0.118$, FDR = 0.0012; T47D: $\Delta PSI = -0.164$, FDR = 8.8×10^{-6}). This splicing change was only partially maintained in MCF7 cells

during repopulation (ΔPSI = −0.097) but was almost fully reversed in T47D cells (ΔPSI = −0.031, FDR = 1.0) (Figure 20).

The fact that SRSF1 is involved in the TIS context is particularly pertinent given the work of Bajtai et al. [15], who identified a number of spliceosome-associated proteins as being enriched in TIS cells and discussed the possibility of stress-induced redistribution and extracellular release of splicing factors (including SRSF1) as a mechanism of cellular adaptation to chemotherapy. Data build on this by showing that SRSF1 itself is subject to major transcriptional and alternative splicing changes during TIS. The different splicing profiles seen between MCF7 and T47D also point to the possibility of cell line-specific regulation of SRSF1 during TIS and REPOP. The functional implications of the observed exon skipping event (in terms of its potential effect on SRSF1 protein levels or activity), however, remain to be established experimentally.

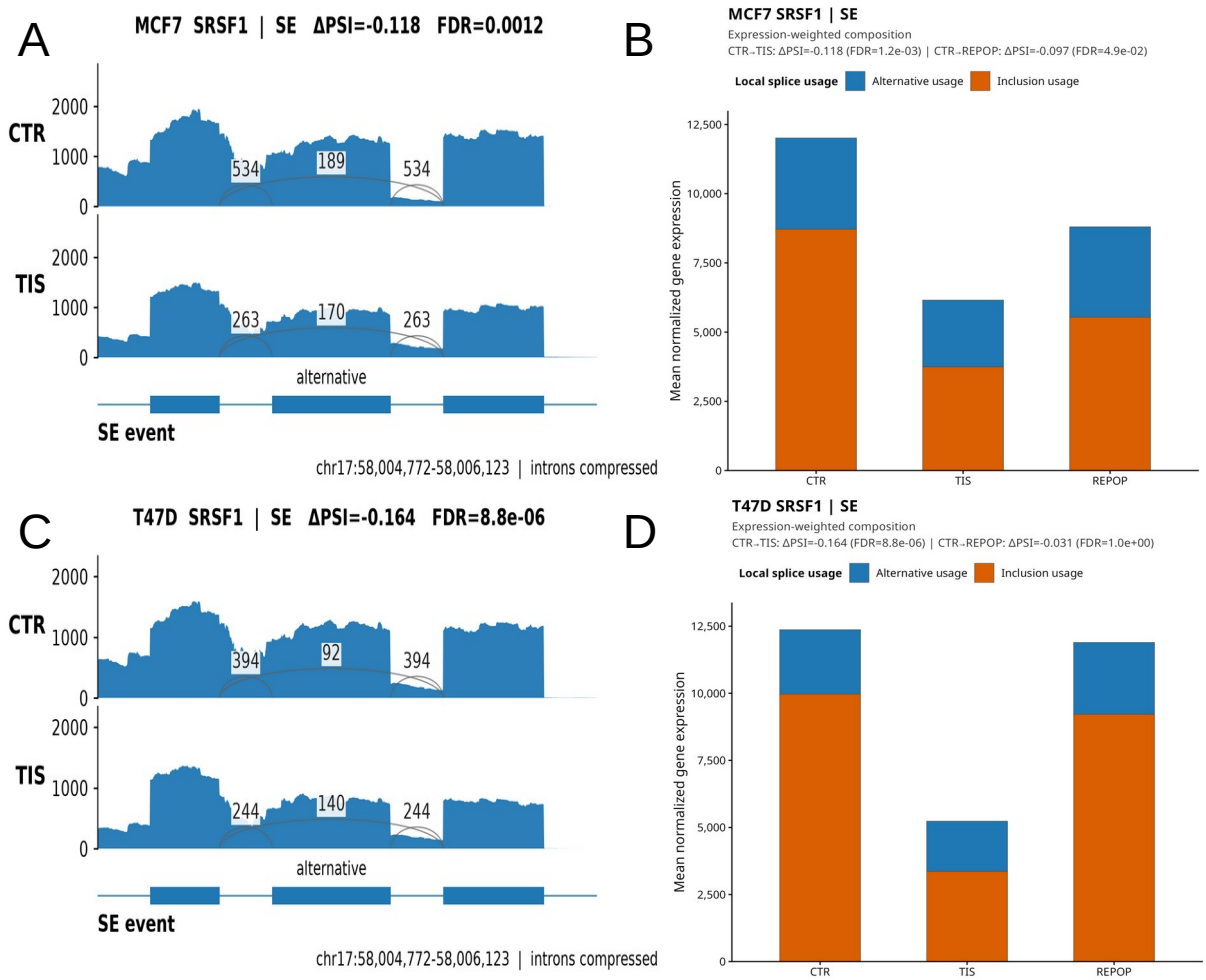


Figure 20. Representative Sashimi plots (A, C) and expression-weighted splice compositions (B, D) of the SRSF1 gene in MCF7 (A–B) and T47D (C–D) cells. The A and C panels show read coverage and splice junctions for control (CTR) and therapy-induced senescence (TIS) states across an alternative skipped exon (SE) event, while the B and D bar plots display the corresponding shifts in local splice usage across the control (CTR), therapy-induced senescence (TIS), and repopulation (REPOP) continuum.

6. Discussion

In this study, I examined how alternative splicing changes as MCF7 and T47D cells undergo therapy-induced senescence (TIS) and then repopulate (REPOP), and how these changes relate to shifts in gene-expression. The key finding is that TIS is associated with extensive splicing remodelling, most of which occurs in genes that show no change in total expression. The REPOP state does not simply revert to the control state: many splicing changes are reversed, some are maintained, or even progress, and a whole new set of splicing changes appears only after the cells have repopulated.

The expression data align with the classic picture of TIS. The samples are clearly separated along the TIS axis, with strong suppression of cell cycle, DNA replication, and DNA repair genes and strong induction of inflammatory, adhesion, and ECM genes, as expected for a senescent secretory phenotype [6,9,10]. The REPOP cells are clearly closer to the controls than the TIS cells are, but they still form a distinct cluster on PC2. This aligns with the original report that the repopulating cells are more similar to the parental cells than to the TIS cells [15] and confirms that the similarity is incomplete. My pipeline (RSEM and DESeq2) reproduces the main expression and pathway patterns of the original analysis, further underscoring the robustness of this transcriptional profile.

The splicing changes should not be interpreted as a wholesale failure of the splicing machinery. The ΔPSI distributions were symmetric and centred on zero for all five event types in both cell lines, and the retained introns showed no systematic trend toward increased retention. This is interesting given that widespread intron retention has been reported in cancer [26] and that senescent cells generally show reduced splicing factor expression [25]. GSEA did show a coordinated downregulation of spliceosome and mRNA processing genes during TIS, which was reversed in REPOP, though the splicing output was selective rather than globally disrupted. The shared splicing changes were indeed enriched in RNA processing and splicing genes, and SRSF1 itself was altered in both expression and splicing during TIS. Splicing factors are known to regulate their own splicing and each other's [29,30], so the splicing machinery could well be both a target and a component of the TIS-associated state. That said, the enrichment is just that, an enrichment. My data do not tell us which splicing factors drive the individual events, and there is a bias toward genes with many exons and high expression, which will also be easier to detect and will lead to over-representation of these categories.

Splicing and differential expression were largely non-overlapping at the gene-significance level. Only 14.7% (MCF7) and 7.1% (T47D) of the consensus DAS genes were also differentially expressed, and ΔPSI spanned the full range of log2 fold changes. Several examples include PSMB3 and ZNF207, which showed substantial shifts in exon usage with stable total expression, and RPS24, which showed the same direction of exon inclusion in both cell lines despite opposite expression responses. This again supports the idea that splicing is a regulatory layer that is in part independent of gene-level transcription [19-22] and that a gene-level analysis alone would miss the majority of these changes. The functional

consequences of the individual events (on transcript stability, coding potential or protein activity) were not assessed here, so they remain entirely untested hypotheses.

The trajectory analysis is the key result for repopulation. Transient and partial trajectories were by far the most common (together constituting around 63% of the TIS-dependent events in MCF7 and over 83% in T47D), which is consistent with the idea that TIS is a reversible state [15,38]. A significant proportion of events were also persistent (26.9% in MCF7, 14.4% in T47D), a small number of which progressed, and a large number were REPOP-specific (1262 in MCF7 versus 122 in T47D). Repopulating cells are therefore not simply a return to the parental splicing profile but represent a distinct state with its own post-transcriptional signature, consistent with the suggestion that escape from senescence involves more than just reversal of the senescent state [14,16,36]. Two caveats are worth noting. The classes are based on the ΔPSI point estimates from 3 replicates, so a non-significant difference between REPOP and CTR is not necessarily evidence of identity, and the 'reversed' class in particular is very sensitive to small initial changes. Also, the bulk data cannot tell us whether a change occurs within the individual cell or reflects a change in population composition (for example, selective survival of a subpopulation during the repopulation phase) [10,13]. I therefore cannot claim that TIS is molecularly remembered in individual cells or their progeny.

The two cell lines shared a core response, but the extent of the response clearly differed. The overlap of significant events was very low (corrected Jaccard index 12.4% for rMATS and 25.7% for LeafCutter at the gene level), and of the high-confidence events that were common to both cell lines, only around 69-70% agreed in direction of change (with only moderate correlation of ΔPSI $r=0.35$-0.41, $\rho=0.43$-0.44). So some changes appear to be core to TIS, while a significant proportion are cell line specific. This makes sense given the molecular differences between MCF7 and T47D [39], which are also apparent in the expression data (the interferon-related signature that was so dominant in TIS in T47D was almost completely switched off upon repopulation in that cell line, whereas MCF7 retained a lot more of the differences). In contrast, the mean absolute ΔPSI profiles of the common events were very similar between the two cell lines within each trajectory class. The cell lines thus appear to differ primarily in the number of events in each class rather than in the magnitude of change within a class. The higher number of significant events in MCF7 should not be interpreted as a more robust biological response, as read depth, variability, and statistical power all affect detection [40].

LeafCutter actually called more significant genes than rMATS, and only 65 genes (in MCF7) and 38 genes (in T47D) were significant in all six of the tool comparison combinations. This is to be expected, as rMATS is testing against a set of pre-defined event types, whereas LeafCutter is testing intron usage within clusters [31,35]. Crucially, when both tools could quantify the same event, they agreed on the direction in 98% of the unfiltered matched cases. So they primarily disagree over which events are significant rather than the direction of change. This justifies focusing on events supported by both and being cautious

about gene-level overlap, since the same gene can be called significant through different introns.

The study has several limitations. It is bulk RNA-seq with three replicates per condition from two ER+ luminal cell lines, so it is unlikely to be generalisable to other subtypes or to tumors *in vivo*. The short-read data can only report on relative exon/intron usage and do not provide full-length isoforms or protein levels [33]. It is an observational study, and there is no evidence of causality between any splicing changes and senescence maintenance, escape from senescence, or subsequent proliferation. The obvious next step would be to validate the selected events by RT-PCR, then move to isoform-resolved long-read sequencing, use single-cell data to resolve heterogeneity, and perturb the candidate regulators (e.g., SRSF1). The most interesting candidates are the persistent and REPOP-specific events common to both cell lines.

7. Summary, further plans

Therapy-induced senescence in MCF7 and T47D cells is accompanied by extensive remodelling of alternative splicing in addition to the well-documented transcriptional response. Most of the splicing changes occur in genes without significant differential expression, and the CTR–TIS–REPOP trajectories show that they are reversed only in part: some are transient, while others persist, progress or arise only after repopulation. A subset of events is supported by both rMATS and LeafCutter and shows the same direction in both cell lines, although the magnitude of the response is cell line-dependent. Genes with shared splicing changes are enriched for RNA processing, cell cycle and DNA repair functions, and the splicing factor SRSF1 itself changes in both expression and splicing during TIS. This suggests that the RNA processing machinery may be part of the TIS-associated state. Functional studies are needed to show whether these events help maintain senescence or enable escape from it.

My next steps are to extend the alternative splicing analysis to characterize the identified events in greater detail and investigate their potential biological relevance during senescence and repopulation. In particular, I would like to further examine the candidate splicing events and their behavior across different cell states to see whether I can identify any events associated with senescence escape. Additionally, I plan to analyze single-cell RNA sequencing data to investigate the expression patterns of selected candidate genes at single-cell resolution. This would allow the determination of which cell states or subpopulations are associated with the observed expression changes, and whether any candidates are enriched in senescent or repopulating cells. Integrating this with the bulk RNA-seq data would provide a much more detailed picture of the molecular changes occurring during the transition from senescence to repopulation, thereby helping to identify strong candidates for further functional work.

8. Acknowledgements

First and foremost, I would like to express my deepest gratitude to my external supervisor, Csaba Kiss, for his continuous guidance, patience, and unwavering support. Without his invaluable help and mentorship, I would not have been able to complete this thesis.

I am also profoundly thankful to my internal supervisor, Dr. János Juhász, for his valuable advice, feedback, and support during the preparation of this work.

My sincere thanks go to Dr. Sándor Spisák for enthusiastically welcoming me into his laboratory and providing an inspiring, supportive environment where I could conduct my research.

Finally, I'm eternally grateful to my family for their unconditional love, patience, and support. A special thanks to my sister, who was always there to listen to my endless ramblings and train of thought over the years. Their belief in me has been the single biggest source of strength for me throughout this whole thing.

9. Data availability

The bulk RNA-seq datasets are available in the NCBI GEO database under the accession number GSE287953. Analysis code has been uploaded to GitHub (https://github.com/sonkolyakos/TDK)

10. Citations

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